MIE286 Lecture Notes After Midterm

Probability and Statistics

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Chapter 1

Chapter 8: Fundamental Sampling Distributions and Data Descriptions

1.1 Key Definitions

Population: The totality of observations with which we are concerned. In other words, it is the entire group of individuals or measurements of interest. We often cannot observe the entire population, so we rely on samples to make inferences.

Sample: A subset of a population. We collect data on the sample and use it to draw conclusions about the population.

Parameter: A number that describes the population. Parameters are fixed but in practice unknown. For example, the population mean μ and variance σ^2 are parameters. Since we rarely observe the whole population, we estimate parameters using statistics.

Statistic: A function of sample data. It is itself a random variable with a distribution. For example, the sample mean $\bar{X}$ and sample variance S^2 are statistics. Because a statistic depends on which sample happens to be drawn, different samples produce different values of the statistic.

The key insight is that a statistic is a random variable. This means we can talk about its probability distribution, which leads to the concept of a sampling distribution.

1.2 Random Samples

Definition: Simple Random Sample

The random variables $X_1, X_2, \ldots, X_n$ are said to form a **(simple) random sample** of size n if the X_i's are **i.i.d.** (independent, identically distributed), with each X_i having probability distribution $f(x)$.

Because the X_i are independent and identically distributed, the joint distribution factors as:

$$f(x_1, x_2, \ldots, x_n) = f(x_1)\, f(x_2) \cdots f(x_n).$$

Why independence matters: Independence means that the value of one observation does not affect the probability of any other observation. This is a reasonable assumption when the population is very large relative to the sample, or when sampling is done with replacement. In practice, even sampling without replacement from a large population is approximately independent.

The sample mean is denoted $\bar{X}$ and computed as:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

Conceptual picture: Imagine taking many samples of size n from the population. From each sample, compute $\bar{X}$. Each sample gives a different $\bar{X}$. The collection of all these sample means forms a distribution.

X_1	X_{11}	X_{21}	$\cdots$	$X_1 \sim f(x)$
$\vdots$	$\vdots$	$\vdots$		
X_n	X_{1n}	X_{2n}		$X_n \sim f(x)$
$\bar{X}$	$\bar{X}_1$	$\bar{X}_2$		$\bar{X}$ is a r.v. $\longrightarrow$ Sampling Distribution

Each column represents a separate sample drawn from the same population. The bottom row shows the sample mean computed for each sample. The distribution of all these sample means is the *sampling distribution* of $\bar{X}$.

1.3 Sampling Distribution

Definition: Sampling Distribution

The probability distribution of a statistic is called a **sampling distribution**. It describes the variability of the statistic across all possible samples of the same size from the same population.

The sampling distribution depends on:

- the size of the population,
- the size of the sample, and
- the method of sampling.

Why sampling distributions matter: The sampling distribution tells us how much a statistic (like $\bar{X}$) would vary from sample to sample. This is the foundation for making inferences about population parameters—it allows us to quantify uncertainty.

1.4 Sampling from a Normal Distribution

1.4.1 Theorem: Distribution of the Sample Mean from a Normal Population

Theorem (Normal Population)

Suppose a random sample of size n is taken from $N(\mu, \sigma^2)$. That is,

$$X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu,\, \sigma^2).$$

Each observation X_i, $i = 1, 2, \ldots, n$, has the same distribution. Then:

$$\bar{X} \sim N\left(\mu,\ \frac{\sigma^2}{n}\right)$$

and equivalently,

$$\sum_{i=1}^{n} X_i \sim N(n\mu,\ n\sigma^2).$$

This was shown through moment generating functions (mgf's). Recall that the mgf of $N(\mu, \sigma^2)$ is $M_X(t) = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right)$.

Proof (via mgf's):

The mgf of a single $X_i \sim N(\mu, \sigma^2)$ is:

$$M_{X_i}(t) = \exp\left(\mu t + \tfrac{1}{2}\sigma^2 t^2\right).$$

For the sum $\sum_{i=1}^{n} X_i$, since the X_i are independent:

$$M_{\sum X_i}(t) = \prod_{i=1}^{n} M_{X_i}(t) = \left[\exp\left(\mu t + \tfrac{1}{2}\sigma^2 t^2\right)\right]^n = \exp\left(n\mu t + \tfrac{1}{2}\, n\sigma^2 t^2\right).$$

This is the mgf of $N(n\mu,\ n\sigma^2)$, so $\sum_{i=1}^{n} X_i \sim N(n\mu,\ n\sigma^2)$.

For the sample mean $\bar{X} = \frac{1}{n}\sum X_i$, we use $M_{\bar{X}}(t) = M_{\sum X_i}(t/n)$:

$$M_{\bar{X}}(t) = \exp\left(n\mu \cdot \frac{t}{n} + \frac{1}{2}\, n\sigma^2 \cdot \frac{t^2}{n^2}\right) = \exp\left(\mu t + \frac{1}{2} \cdot \frac{\sigma^2}{n} \cdot t^2\right).$$

This is the mgf of $N\left(\mu, \frac{\sigma^2}{n}\right)$. Therefore $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$. □

Key takeaway: The sample mean $\bar{X}$ is centered at the true population mean μ. Its variance σ^2/n shrinks as

n grows, meaning larger samples produce more precise estimates of μ.

1.5 The Central Limit Theorem (CLT)

The previous result required the population to be normal. The Central Limit Theorem is a much more powerful result that applies to *any* distribution.

Central Limit Theorem

Let $X_1, X_2, \ldots, X_n$ be independent random variables from the *same* distribution with mean μ and variance σ^2. Then the limiting form of the distribution of

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}},$$

as $n \to \infty$, is the standard normal distribution.

Rule of thumb: The approximation is generally considered adequate when $n \geq 30$.

Professor's remark: "The CLT is the most important theorem in this course. It is what allows us to do all of statistics — without the CLT, we couldn't build confidence intervals or do hypothesis tests unless we knew the exact population distribution. Remember: the CLT applies to the **sample mean**, not to the raw data. A large sample does NOT make the data itself normal."

Why this is powerful: Even if the individual X_i follow a skewed, uniform, exponential, or other non-normal distribution, once we average enough of them together, the distribution of the average becomes approximately normal. This is the theoretical foundation for much of classical statistics.

Equivalent formulations of the CLT:

1. If X_i are i.i.d. from a distribution with mean μ and variance σ^2, then:

$$\bar{X} \overset{\text{approx}}{\sim} N\left(\mu,\ \frac{\sigma^2}{n}\right) \qquad \text{for large } n.$$

2. For the sum: if $\bar{X} \overset{\text{approx}}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right)$, then:

$$\sum_{i=1}^{n} X_i \overset{\text{approx}}{\sim} N(n\mu,\ n\sigma^2).$$

Remark: $n \geq 30$ is a good cutoff for the approximation. For populations that are already close to normal, the CLT kicks in for much smaller n. For highly skewed populations, you may need larger n.

1.6 The t-Distribution (Student's t)

The CLT and the Z-statistic require us to know the population standard deviation σ. In practice, σ is almost never known. When we replace σ with the sample standard deviation S, the resulting statistic no longer

follows a normal distribution — it follows a **t-distribution**.

1.6.1 Why We Need the t-Distribution

Recall from the CLT that when σ is known:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1).$$

But if σ is unknown, we estimate it with S and form:

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}.$$

This T statistic has *more variability* than Z because S itself is random — it changes from sample to sample. The extra uncertainty (especially when n is small) means the tails are heavier than a standard normal. This is exactly what the t-distribution captures.

1.6.2 Definition and Theorem

Definition: t-Distribution

If $Z \sim N(0,1)$ and $V \sim \chi^2(\nu)$ are **independent**, then:

$$T = \frac{Z}{\sqrt{V/\nu}} \sim t(\nu),$$

where ν is called the **degrees of freedom**.

Theorem: One-Sample t-Statistic

Let $X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$ where both μ and σ^2 are **unknown**. Then:

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1).$$

This follows because $\bar{X}$ and S^2 are independent (a special property of the normal distribution), and $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$.

Key properties of $t(\nu)$:

- Symmetric about 0 and bell-shaped (like the normal), but with **heavier tails**.
- As $\nu \to \infty$, $t(\nu) \to N(0,1)$. In practice, for $\nu \geq 30$ the t and Z distributions are very similar.
- $E(T) = 0$ (for $\nu > 1$) and $\text{Var}(T) = \dfrac{\nu}{\nu - 2}$ (for $\nu > 2$), which is always greater than 1.

Professor's remark: "Use Z when σ is known or n is large. Use t when σ is unknown and n is small. On the exam, the problem will tell you whether σ is known or not — read carefully. If it says '$\sigma = \ldots$' you use Z; if it says '$s = \ldots$' (sample standard deviation), you use t."

1.6.3 When to Use Z vs. t

Scenario	Statistic	Distribution
σ known, any n, normal pop.	$Z = \dfrac{\bar{X} - \mu}{\sigma/\sqrt{n}}$	$N(0,1)$
σ unknown, normal pop.	$T = \dfrac{\bar{X} - \mu}{S/\sqrt{n}}$	$t(n-1)$
σ known or unknown, $n \geq 30$	either	$\approx N(0,1)$ by CLT

1.6.4 Confidence Intervals Using the t-Distribution

Confidence Interval for μ (σ Unknown)

A $(1-\alpha) \times 100\%$ confidence interval for μ when σ is unknown and the population is normal (or n is large) is:

$$\bar{X} \pm t_{n-1;\,\alpha/2} \frac{S}{\sqrt{n}},$$

where $t_{n-1;\,\alpha/2}$ is the critical value from the t-distribution with $n-1$ degrees of freedom such that $P(T > t_{n-1;\,\alpha/2}) = \alpha/2$.

Comparison with the Z-interval: When σ is known, the CI is $\bar{X} \pm z_{\alpha/2} \dfrac{\sigma}{\sqrt{n}}$. The t-interval replaces σ with S and $z_{\alpha/2}$ with $t_{n-1;\,\alpha/2}$. Because t critical values are slightly larger than z critical values (especially for small n), the t-interval is **wider**, reflecting the extra uncertainty from estimating σ.

Key takeaway: The t-distribution is what bridges the gap between theory (where σ is known) and practice (where σ must be estimated). Almost every real-world confidence interval and hypothesis test for the mean uses the t-distribution.

1.7 Data Descriptions: Quantiles, Boxplots, and QQ-Plots

1.7.1 Quantiles

A quantile of a sample (say x_p) is the value below which a specific fraction p of the data values fall. Quantiles are often described as **percentiles**.

Quartiles are special cases of quantiles:

- 1st quartile Q_1 (25th percentile),
- 2nd quartile Q_2 (50th percentile, i.e., the median),
- 3rd quartile Q_3 (75th percentile).

The **five-number summary** of a dataset consists of: min, Q_1, Q_2, Q_3, max.

1.7.2 Boxplot

A boxplot is a graphical display based on the five-number summary. It shows:

- a box from Q_1 to Q_3 (the **interquartile range**, $\text{IQR} = Q_3 - Q_1$),
- a line at the median inside the box,
- whiskers extending to the smallest and largest observations within $1.5 \times \text{IQR}$ of the quartiles,
- individual points (possible outliers) beyond the whiskers.

Understanding the Interquartile Range (IQR)

The **Interquartile Range (IQR)** is one of the most important measures of spread:

$$\boxed{\text{IQR} = Q_3 - Q_1}$$

What it captures: The IQR represents the range of the *middle 50%* of the data. Unlike the full range ($\max - \min$), the IQR is **resistant to outliers** because it ignores the extreme values in the tails.

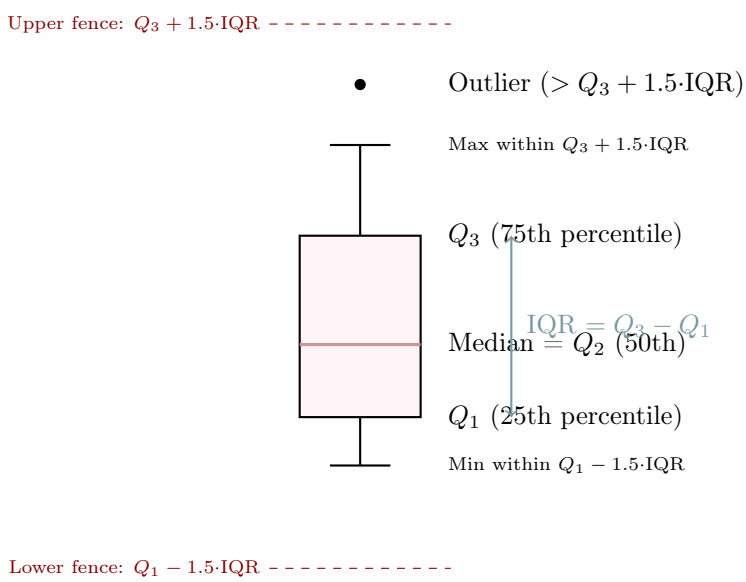
How to compute Q_1 and Q_3:

1. Sort the data in ascending order.
2. Find the median (Q_2): this splits the data into a lower half and an upper half.
3. Q_1 = median of the **lower half** of the data (below the overall median).
4. Q_3 = median of the **upper half** of the data (above the overall median).

Outlier detection using IQR:

- **Lower fence:** $Q_1 - 1.5 \times \text{IQR}$
- **Upper fence:** $Q_3 + 1.5 \times \text{IQR}$
- Any data point below the lower fence or above the upper fence is flagged as a **possible outlier**.
- The whiskers of the boxplot extend only to the most extreme data points *within* these fences, not to the fences themselves.

Professor's remark: "The IQR is preferred over the range or standard deviation when you suspect outliers in the data, because it only looks at the middle half. This makes the boxplot a robust summary."



1.7.3 Quantile Plot

A quantile plot displays data values on the vertical axis against the fraction of observations exceeded by the data value on the horizontal axis. Key properties:

- All data points are plotted (unlike a boxplot which summarizes).
- It is **descriptive**, not diagnostic.

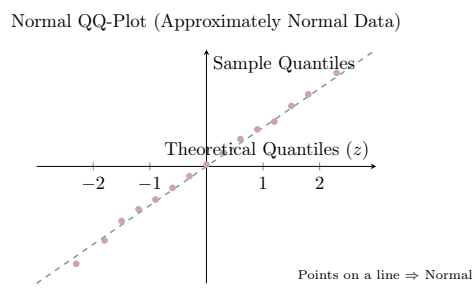
1.7.4 Normal Quantile-Quantile (QQ) Plot

The normal QQ-plot is a **diagnostic** tool used to determine whether a data set came from a normal distribution.

- Plot the data quantiles (vertical axis) against the corresponding quantiles of the standard normal distribution (horizontal axis).
- If the data are approximately normal, the points will fall approximately on a **straight line**.
- Systematic deviations from linearity indicate non-normality.
- **Outliers** appear as points that are far from the overall pattern.

Note on the textbook: The x-axis labels in the book's QQ-plot figure should be $-2, -1, 0, 1, 2$ (standard normal quantiles), not the values shown.

Professor's remark: "On the exam, you will be asked to interpret QQ-plots, not to construct them from scratch. The key patterns to know: if the points curve upward at the right end, the data are right-skewed; if they curve downward at the left end, the data are left-skewed; if they form an S-shape, the data have heavier tails than normal. A straight line means approximately normal."



1.7.5 Normal Probability Plot

Similar to the QQ-plot, but rather than corresponding z-scores we have corresponding standard normal percentiles and a special scale. The interpretation is the same:

- Look for a straight line — systematic deviations indicate non-normality.
- Outliers appear as points far from the overall pattern.

1.8 Chebyshev's Inequality

Theorem: Chebyshev's Inequality

The probability that X will assume a value within k standard deviations of the mean is at least $1-\frac{1}{k^2}$:

$$P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2}.$$

Interpretation: Chebyshev's inequality gives a lower bound on probability that works for **any** distribution (as long as the mean and variance exist). It does not require normality.

Proof (continuous case):

We start with the definition of variance:

$$\sigma^2 = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\, dx.$$

Split the integral into three regions:

$$\sigma^2 = \int_{-\infty}^{\mu-k\sigma} (x-\mu)^2 f(x)\, dx + \int_{\mu-k\sigma}^{\mu+k\sigma} (x-\mu)^2 f(x)\, dx + \int_{\mu+k\sigma}^{\infty} (x-\mu)^2 f(x)\, dx.$$

Since the middle integral is non-negative, we can drop it to get a lower bound:

$$\sigma^2 \geq \int_{-\infty}^{\mu-k\sigma} (x-\mu)^2 f(x)\, dx + \int_{\mu+k\sigma}^{\infty} (x-\mu)^2 f(x)\, dx.$$

For $x \leq \mu - k\sigma$ or $x \geq \mu + k\sigma$, we have $|x - \mu| \geq k\sigma$, so $(x-\mu)^2 \geq k^2\sigma^2$. Substituting:

$$\sigma^2 \geq k^2\sigma^2 \left[\int_{-\infty}^{\mu - k\sigma} f(x)\,dx + \int_{\mu + k\sigma}^{\infty} f(x)\,dx \right].$$

Dividing both sides by $k^2\sigma^2$:

$$\int_{-\infty}^{\mu - k\sigma} f(x)\,dx + \int_{\mu + k\sigma}^{\infty} f(x)\,dx \leq \frac{1}{k^2}.$$

Therefore:

$$P(\mu - k\sigma < X < \mu + k\sigma) = 1 - \left[\int_{-\infty}^{\mu - k\sigma} f(x)\,dx + \int_{\mu + k\sigma}^{\infty} f(x)\,dx \right] \geq 1 - \frac{1}{k^2}. \qquad \square$$

1.9 R Code: QQ-Plots and CLT Demonstration

1.9.1 Creating Random Samples and QQ-Plots

```
# Creating random samples from a given distribution
x <- rf(2000, 3, 27)       # random sample of size 2000 from F(3, 27)
x <- rnorm(100, 2, 100)    # random sample of size 100 from N(2, 100)

# Plotting a normal quantile-quantile plot
qqnorm(x)
qqline(x)

# For the normal probability plot (requires "e1071" package)
library(e1071)
probplot(x)    # draw a normal probability plot
```

1.9.2 Demonstrating the CLT

The idea is to:

1. Generate random data from a non-normal distribution (e.g., $F(3, 27)$, which is right-skewed).
2. Repeatedly draw samples of size n and compute the sample mean each time.
3. Plot the distribution of these sample means.
4. Observe that as n increases, the distribution of $\bar{X}$ becomes more and more normal.

```
# Show that the F(3,27) distribution is right-skewed
x <- rf(2000, 3, 27)
hist(x)

# CLT demonstration
```

```
n <- 4     # try different values: n=4, 10, 30, 100
xbar <- array(0, dim = c(0, 1000))  # vector of 1000 zeros

for(i in 1:1000) {
  xbar[i] <- mean(rf(n, 3, 27))      # sample mean of n obs from F(3,27)
}

hist(xbar)        # should look more normal as n increases
qqnorm(xbar)      # should fall on a straight line
qqline(xbar)
```

Key observation: When $n = 4$, the histogram of $\bar{X}$ is still somewhat skewed. As n increases to 30 or more, the histogram looks approximately normal and the QQ-plot becomes nearly linear. This is the Central Limit Theorem in action.

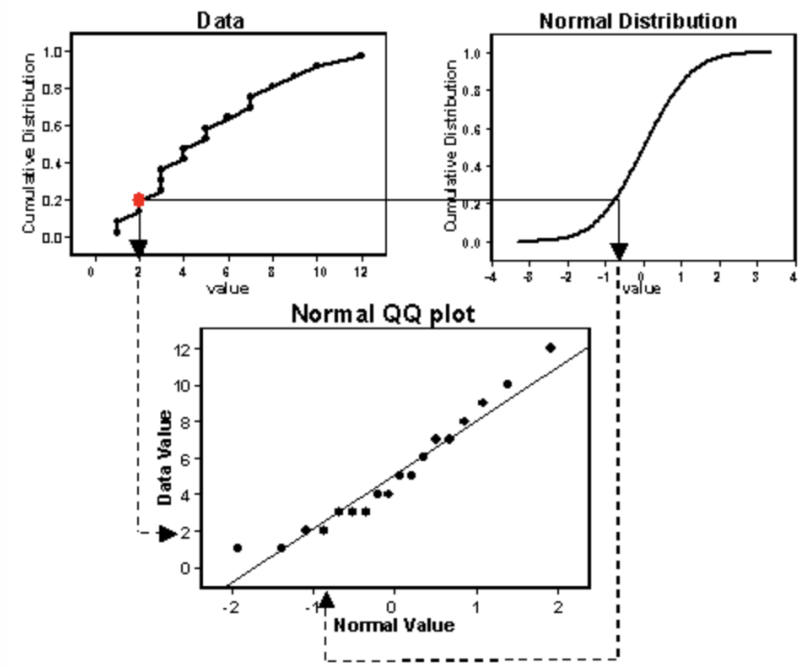


Figure 1.1: Plots example

1.10 True or False: Conceptual Questions

The following true/false questions test understanding of sampling distributions and the CLT:

1. **As n (sample size) increases, the mean of the sampling distribution of $\bar{X}$ decreases.**

   False. The mean of $\bar{X}$ is always μ, regardless of n. The sample size does not affect the center of the sampling distribution.

2. **As n increases, the standard deviation of the sampling distribution of $\bar{X}$ decreases.**

True. The standard deviation of $\bar{X}$ is $\sigma/\sqrt{n}$, which decreases as n grows. Larger samples produce more precise estimates.

3. **The mean $\bar{X}$ of a random sample of size 4 from a left-skewed distribution is approximately normally distributed.**

   False. The CLT requires $n \geq 30$ (approximately) for the normal approximation to be adequate. With $n = 4$ from a skewed distribution, $\bar{X}$ will not be approximately normal.

4. **The distribution of the proportion of successes $\hat{p}$ in a sufficiently large sample is approximately normal with mean p and std. dev. $\sqrt{np(1-p)}$.**

   False. The standard deviation of $\hat{p} = X/n$ is $\sqrt{p(1-p)/n}$, not $\sqrt{np(1-p)}$. The quantity $\sqrt{np(1-p)}$ is the standard deviation of the *count* X, not the proportion $\hat{p}$.

5. **If $\bar{X}$ is the mean of a simple random sample of size 9 from $N(500, \sigma^2 = 18)$, then $\bar{X}$ has a normal distribution with mean 500 and variance 36.**

   False. Since $X_i \sim N(500, 18)$, we have $\bar{X} \sim N(500, 18/9) = N(500, 2)$. The variance of $\bar{X}$ is $18/9 = 2$, not 36.

6. **A large sample from a skewed population distribution will have an approximately normal histogram.**

   False. The *data itself* will still look skewed. It is the *sampling distribution of* $\bar{X}$ that becomes approximately normal, not the distribution of the raw data.

7. **The mean of a population will be normally distributed if the population is quite large.**

   False. The population mean μ is a fixed number (a parameter), not a random variable. It does not have a distribution. The CLT applies to the *sample mean* $\bar{X}$, not to the population mean.

8. **The average blood cholesterol level recorded in a simple random sample of 100 students from a large population will be approximately normally distributed.**

   True. By the CLT, with $n = 100 \geq 30$, the sample mean $\bar{X}$ is approximately normal regardless of the shape of the population distribution.

1.11 Examples: Applying the CLT to the Sum of Random Variables

1.11.1 Example 1: Airline Passenger Weights

Problem. Weights of airline passengers are known to have a distribution with mean $\mu = 75\,\text{kg}$ and standard deviation $\sigma = 10\,\text{kg}$. A certain plane has a passenger weight capacity of $7700\,\text{kg}$. What is the probability that a flight of 100 passengers will exceed the capacity?

Setup. Let X_i denote the weight (in kg) of the i-th passenger, so $X_i \sim (\mu = 75, \sigma = 10)$. Since passengers are randomly selected, $X_1, \ldots, X_{100}$ are i.i.d. We want:

$$P\left(\sum_{i=1}^{100} X_i > 7700\right).$$

Intuition. If each passenger weighs 75 kg on average, then 100 passengers should total around $100 \times 75 = 7500\,\text{kg}$. The question is how likely the total exceeds 7700 kg.

Solution. By the CLT (since $n = 100 \geq 30$):

$$\sum_{i=1}^{100} X_i \overset{\text{approx}}{\sim} N(100 \times 75,\ 100 \times 100) = N(7500,\ 10000)\,.$$

The variance of the sum is $n\sigma^2 = 100 \times 10^2 = 10{,}000$, so the standard deviation is $\sqrt{10{,}000} = 100$.

$$P\left(\sum_{i=1}^{100} X_i > 7700\right) = P\left(Z > \frac{7700 - 7500}{\sqrt{100 \times 100}}\right) = P\left(Z > \frac{200}{100}\right) = P(Z > 2) \approx \boxed{0.0228.}$$

Conclusion: There is approximately a 2.28% chance the flight exceeds the weight capacity.

1.12 Inferences on the Population Mean

So far, we have used probability to compute things given known parameters (e.g., "given $\mu = 75$, find $P(\ldots)$"). Now we flip the question: given observed data, what can we infer about the unknown μ?

1.12.1 The Setup

We have a sample $X_1, \ldots, X_n$ that are i.i.d. with unknown mean μ and known (or estimable) standard deviation σ. We observe the sample mean $\bar{X}$ and ask: is the data consistent with a particular assumed value of μ?

By the CLT (when n is large), we know:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

1.12.2 Example 2: Cylindrical Component Diameters

Problem. A manufacturing process produces cylindrical components. An engineer conjectures that the population mean diameter is $\mu = 5\,\text{mm}$. A random sample of $n = 100$ parts is selected, yielding $\bar{x} = 5.027\,\text{mm}$. It is known that $\sigma = 0.1\,\text{mm}$. Does the sample support the engineer's conjecture?

Note: In practice, σ is rarely "known" exactly. Here we take it as a given assumption — perhaps it was established from extensive prior production data. The case where σ must be estimated from the sample leads to t-procedures covered later.

Setup. Let X_i be the diameter of the i-th component (in mm). Then $X_1, \ldots, X_{100}$ are i.i.d. with unknown μ and $\sigma = 0.1$. By the CLT:

$$\bar{X} \sim N\left(\mu,\ \frac{0.1^2}{100}\right) = N\left(\mu,\ \frac{0.01}{100}\right).$$

The standard deviation of $\bar{X}$ is $\dfrac{0.1}{\sqrt{100}} = 0.01$.

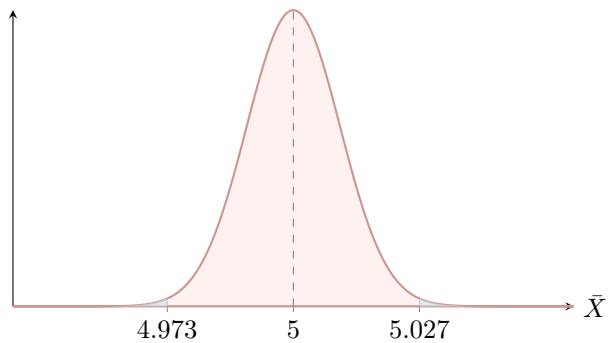
Assuming $\mu = 5$. How unlikely is it to observe a sample mean that deviates from 5 by 0.027 mm or more?

$$P\Big(\bar{X} \le 4.973 \;\text{ or }\; \bar{X} \ge 5.027 \;\Big|\; \mu = 5\Big) = 1 - P(4.973 < \bar{X} < 5.027)$$

By symmetry:

$$= 2\,P\left(Z > \frac{5.027 - 5}{0.1/10}\right) = 2\,P\left(Z > \frac{0.027}{0.01}\right) = 2\,P(Z > 2.7) \approx \boxed{0.007.}$$

Visualization. The sampling distribution of $\bar{X}$ under $\mu = 5$ is shown below. The shaded tails represent the probability of observing a sample mean as extreme as or more extreme than $\bar{x} = 5.027$.



Interpretation. A probability of $0.007 = 0.7\%$ means that, *if the true mean really were 5 mm*, observing a sample mean as far as 5.027 mm from 5 would be a very rare event. Since 0.007 is well below the conventional threshold of $\alpha = 0.05$, we say the observed data **does NOT support** the engineer's conjecture that $\mu = 5\,\text{mm}$.

On the significance threshold $\alpha = 0.05$: In science and engineering, the conventional cutoff for declaring a result statistically significant is $\alpha = 0.05$ (5%). This is sometimes called the "risk bound" or significance level. If the probability of seeing data at least as extreme as what we observed—assuming the null hypothesis is true—falls below α, we reject the null hypothesis. Here, $0.007 < 0.05$, so we reject $\mu = 5$.

Professor's remark: "The p-value is NOT the probability that H_0 is true. It is the probability of seeing data at least as extreme as what we observed, **given that H_0 is true**. Small p-value means the data are surprising under H_0, so we reject H_0. Think of it like a rare event — if something very unlikely happened under the assumption, maybe the assumption is wrong."

1.13 Sampling Distribution of the Difference Between Two Averages

Theorem: Two-Sample Difference of Means

Let independent random samples of sizes n_1 and n_2 be drawn from two populations (discrete or continuous) with means μ_1, μ_2 and variances σ_1^2, σ_2^2, respectively. If n_1 and n_2 are sufficiently large, then:

$$\bar{X}_1 - \bar{X}_2 \overset{\text{approx}}{\sim} N\left(\mu_1 - \mu_2,\ \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right).$$

Proof sketch. By the CLT applied to each sample independently:

$$\bar{X}_1 \overset{\text{approx}}{\sim} N\left(\mu_1, \frac{\sigma_1^2}{n_1}\right) \quad \text{(from CLT)} \qquad \text{and} \qquad \bar{X}_2 \overset{\text{approx}}{\sim} N\left(\mu_2, \frac{\sigma_2^2}{n_2}\right) \quad \text{(from CLT)}.$$

Since the samples are independent, and from Chapter 7 we know that *linear combinations of independent normal random variables are normal*, we get:

$$\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2,\ \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right). \qquad \square$$

Professor's remark: "When comparing two populations, always set up the problem under H_0. If H_0 says the means are equal, then $\mu_1 - \mu_2 = 0$, and your distribution is centered at 0. The variance is always $\sigma_1^2/n_1 + \sigma_2^2/n_2$ — the variances **add**, they never subtract, because variability always increases when you combine independent sources."

1.13.1 Example 3: Comparing Paint Drying Times

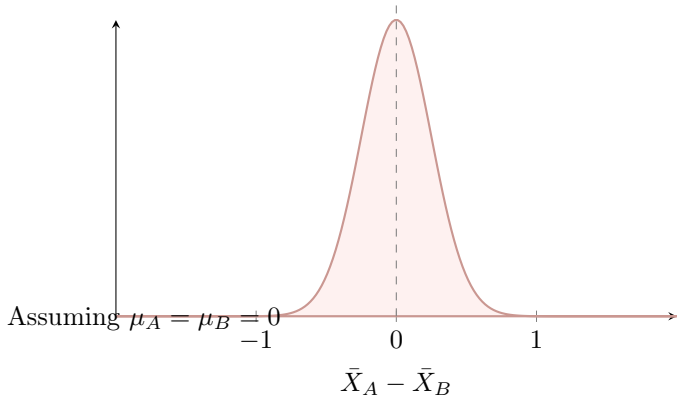
Problem. In a manufacturing study, 32 specimens are painted with Paint A and another 32 are painted with Paint B. The drying time in hours is recorded for each specimen. It is known that $\sigma = 1$ hour for both paints. The observed difference in sample averages was $|\bar{x}_A - \bar{x}_B| = 1$. Is there evidence that $\mu_A \neq \mu_B$?

Setup. Under the assumption that $\mu_A = \mu_B$ (i.e., the paints are equally fast), the sampling distribution of the difference is:

$$\bar{X}_A - \bar{X}_B \overset{\text{approx}}{\sim} N\left(\mu_A - \mu_B,\ \frac{\sigma_A^2}{32} + \frac{\sigma_B^2}{32}\right) = N\left(0,\ \frac{1}{32} + \frac{1}{32}\right) = N\left(0,\ \frac{1}{16}\right).$$

The standard deviation of $\bar{X}_A - \bar{X}_B$ is $\frac{1}{4}$.

Visualization. Under $\mu_A = \mu_B = 0$, the distribution is centered at 0 with the observed difference of ± 1 shown in the tails:



Calculation.

$$P\Big(\bar{X}_A - \bar{X}_B > 1 \;\Big|\; \mu_A = \mu_B\Big) + P\Big(\bar{X}_A - \bar{X}_B < -1 \;\Big|\; \mu_A = \mu_B\Big)$$
$$= 2\,P\left(\frac{(\bar{X}_A - \bar{X}_B) - 0}{1/4} > \frac{1-0}{1/4}\right) = 2\,P(Z > 4) < 0.0001.$$

Conclusion. The probability of observing a difference as large as 1 hour (in absolute value) under the null hypothesis is less than 0.0001. This is far below any reasonable significance level. We **reject** $\mu_A = \mu_B$ and conclude there is strong evidence of a difference in mean drying times.

1.14 Summary: Hypothesis Testing Procedure

Step-by-Step Hypothesis Testing (from the Board)

Step 1: State the hypotheses.

- H_0: Null hypothesis (the "status quo" or "nothing special" claim).
- H_a: Alternative hypothesis (what we suspect or want to show).

Step 2: Compute the test statistic.

- σ known, or large n: $Z = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$
- σ unknown, small n (normal pop.): $T = \dfrac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t(n-1)$
- Two-sample (equal var.): $T = \dfrac{\bar{X}_1 - \bar{X}_2}{S_p\sqrt{1/n_1 + 1/n_2}} \sim t(n_1 + n_2 - 2)$
- Variance test: $F = \dfrac{S_1^2}{S_2^2} \sim F(n_1 - 1, n_2 - 1)$

Step 3: Determine the rejection region OR compute the p-value (The p-value is the probability of getting results as extreme as (or more extreme than) what we observed, assuming H_0 x is true).
Step 4: Make a decision. Reject H_0 if test stat falls in rejection region (or $p < \alpha$).
Step 5: State the conclusion in context.

Professor's remark: "On the exam, always show all five steps. Even if you get the arithmetic wrong, showing the correct procedure earns partial credit. Also remember: the pooled variance formula (s_p^2) is only used when you have evidence that the two population variances are equal — typically from an F-test."

Confidence Intervals vs. Hypothesis Tests

Connection: A two-sided $(1-\alpha)$ confidence interval and a two-sided α-level test are equivalent:

- If μ_0 falls **inside** the CI $\Rightarrow$ fail to reject H_0 at level α.
- If μ_0 falls **outside** the CI $\Rightarrow$ reject H_0 at level α.

This is why you can sometimes answer a hypothesis test question by looking at a confidence interval (as in Practice 8.3b).

1.15 Practice Problems

Practice 8.1 — CLT: Sum of Random Variables (adapted from 2014 exam)

Problem. Suppose that $X_1, \ldots, X_{200}$ are independent continuous random variables with common density function

$$f(x) = 2(1-x) \quad \text{for } 0 \le x \le 1, \quad 0 \text{ otherwise.}$$

Define $S = X_1 + \cdots + X_{200}$.

(a) Compute the mean and variance of X_i.

(b) Using an appropriate method, find an approximation for $P(S > 70)$.

(c) Now define $Y_i = 1$ if $X_i > 0.9$ and $Y_i = 0$ otherwise. Set $T = Y_1 + \cdots + Y_{200}$. Compute $P(T \ge 3)$.

Solution.

(a)

$$E(X) = \int_0^1 x \cdot 2(1-x)\,dx = 2\int_0^1 (x - x^2)\,dx = 2\left[\frac{1}{2} - \frac{1}{3}\right] = \frac{1}{3}.$$

$$E(X^2) = \int_0^1 x^2 \cdot 2(1-x)\,dx = 2\left[\frac{1}{3} - \frac{1}{4}\right] = \frac{1}{6}.$$

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = \frac{1}{6} - \frac{1}{9} = \frac{1}{18}.$$

(b) By the CLT, S is approximately normal with:

$$E(S) = 200 \times \frac{1}{3} = \frac{200}{3} \approx 66.67, \qquad \mathrm{Var}(S) = 200 \times \frac{1}{18} = \frac{200}{18} \approx 11.11.$$

$$P(S > 70) = P\left(Z > \frac{70 - 200/3}{\sqrt{200/18}}\right) = P\left(Z > \frac{70 - 66.67}{\sqrt{11.11}}\right) = P(Z > 1.00) \approx 0.1587.$$

(c) First, find $P(X_i > 0.9)$:

$$P(X_i > 0.9) = \int_{0.9}^1 2(1-x)\,dx = \left[-(1-x)^2\right]_{0.9}^1 = 0 - (-(0.1)^2) = 0.01.$$

So $T = \sum_{i=1}^{200} Y_i \sim \mathrm{Bin}(200,\ 0.01)$.
Exact computation:

$$\begin{aligned} P(T \ge 3) &= 1 - P(T=0) - P(T=1) - P(T=2) \\ &= 1 - \binom{200}{0}(0.01)^0(0.99)^{200} - \binom{200}{1}(0.01)^1(0.99)^{199} - \binom{200}{2}(0.01)^2(0.99)^{198} \\ &\approx 1 - 0.1340 - 0.2707 - 0.2720 = \boxed{0.3233.} \end{aligned}$$

Practice 8.2 — Hypothesis Test on Population Mean (adapted from 2015 exam)

Problem. A program was run 6 times with randomly chosen data sets. The sample mean and sample standard deviation of execution times were $\bar{x} = 228$ ms and $s = 14$ ms, respectively.

(a) Compute a 95% confidence interval for the true mean execution time. State your assumptions.

(b) It is claimed that the overall mean execution time is less than 230 ms. Do the data support this claim? Use $\alpha = 0.05$.

Solution.
(a) The sample is small ($n = 6$) and σ is unknown, so we assume the population is normally distributed and use the t-distribution with $n - 1 = 5$ degrees of freedom.
$t_{5;\,0.025} = 2.571$. The 95% CI is:

$$\bar{x} \pm t_{n-1;\,\alpha/2}\,\frac{s}{\sqrt{n}} = 228 \pm 2.571 \times \frac{14}{\sqrt{6}} = 228 \pm 14.7 = (213.3,\ 242.7)\,\text{ms}.$$

(b) Hypotheses: H_0: $\mu = 230$ vs. H_a: $\mu < 230$.

$$t_{\text{stat}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{228 - 230}{14/\sqrt{6}} = -0.35.$$

The critical value is $-t_{5;\,0.05} = -2.015$. Since $t_{\text{stat}} = -0.35 > -2.015$, we **fail to reject** H_0. The data do not provide sufficient evidence to support the claim.

Practice 8.3 — Two-Sample Inference on Means (adapted from 2015 exam)

Problem. A company supporting two types of computer networks measures downtime (hours/year):

	Type 1	Type 2
Sample size	10	9
Sample mean	83.5	60.4
Standard deviation	34.2	22.3

(a) Test whether the two populations have different variances at $\alpha = 0.10$.

(b) Assuming equal variances, is there a significant difference in mean downtime? Use $\alpha = 0.10$.

Solution.
(a) H_0: $\sigma_1^2 = \sigma_2^2$ vs. H_a: $\sigma_1^2 \neq \sigma_2^2$.

$$F_{\text{stat}} = \frac{s_1^2}{s_2^2} = \frac{34.2^2}{22.3^2} = \frac{1169.64}{497.29} \approx 2.352.$$

Critical value: $F_{(9,8);\,0.05} = 3.39$. Since $2.352 < 3.39$, we **fail to reject** H_0. There is no significant evidence of unequal variances.
(b) Using the pooled t-test (equal variance assumed):

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{9 \times 34.2^2 + 8 \times 22.3^2}{17} \approx 853.24.$$

$t_{17;\,0.05} = 1.740$. The 90% CI for $\mu_1 - \mu_2$:

$$(83.5 - 60.4) \pm 1.740\sqrt{853.24\left(\frac{1}{10} + \frac{1}{9}\right)} = 23.1 \pm 22.85 \approx (0.25,\ 45.95).$$

Since the CI does not contain 0, we **reject** H_0 at $\alpha = 0.10$ and conclude there is a significant difference in mean downtime.

Practice 8.4 — Inference and Power (adapted from 2015 exam)

Problem. Define $X_1, \ldots, X_{100}$ to be independent Normal r.v.'s with unknown mean μ and variance 25. Test H_0: $\mu = 10$ vs. H_1: $\mu \neq 10$ using test statistic $\bar{X} = \frac{1}{100}\sum_{i=1}^{100} X_i$. Reject H_0 if $\bar{X} \geq 10.98$ or $\bar{X} \leq 9.02$.

(a) Find the significance level of the test.

(b) If the true value is $\mu = 10.5$, find the power of the test.

Solution.
(a) Under H_0, $\bar{X} \sim N\big(10,\ \frac{25}{100}\big) = N(10, 0.25)$, so $\mathrm{SD}(\bar{X}) = 0.5$.

$$\alpha = P(\bar{X} \geq 10.98) + P(\bar{X} \leq 9.02) = 2\,P\left(Z > \frac{10.98 - 10}{0.5}\right) = 2\,P(Z > 1.96) = 2(0.025) = \boxed{0.05.}$$

(b) Under $H_1\colon \mu = 10.5$, $\bar{X} \sim N(10.5, 0.25)$.

$$\begin{aligned}
\text{Power} &= P(\bar{X} \geq 10.98 \mid \mu = 10.5) + P(\bar{X} \leq 9.02 \mid \mu = 10.5) \\
&= P\left(Z > \frac{10.98 - 10.5}{0.5}\right) + P\left(Z < \frac{9.02 - 10.5}{0.5}\right) \\
&= P(Z > 0.96) + P(Z < -2.96) \\
&\approx 0.1685 + 0.0015 = \boxed{0.17.}
\end{aligned}$$

Practice 8.5 — Sampling Distribution of Difference (two paints)

Problem. Suppose specimens are painted with Paint A or Paint B. We know $\sigma_A = \sigma_B = 1$ hour, and independent samples of size $n_A = n_B = 32$ are taken. The observed difference is $\bar{x}_A - \bar{x}_B = 1$.

(a) State the null and alternative hypotheses for testing whether the paints differ in mean drying time.

(b) Under H_0, what is the sampling distribution of $\bar{X}_A - \bar{X}_B$?

(c) Compute the p-value and state your conclusion.

Solution.
(a) H_0: $\mu_A = \mu_B$ (equivalently $\mu_A - \mu_B = 0$) vs. H_a: $\mu_A \neq \mu_B$.

(b) Under H_0:

$$\bar{X}_A - \bar{X}_B \sim N\left(0,\ \frac{1}{32} + \frac{1}{32}\right) = N\left(0,\ \frac{1}{16}\right), \quad \text{SD} = \frac{1}{4}.$$

(c)

$$p\text{-value} = 2\,P\big(\bar{X}_A - \bar{X}_B > 1\big) = 2\,P\left(Z > \frac{1-0}{1/4}\right) = 2\,P(Z > 4) < 0.0001.$$

Since $p < 0.05$, we **reject** H_0 and conclude that the paints have different mean drying times.

Practice 8.6 — Hypothesis Test: Binomial with Normal Approximation (adapted from 2014 exam)

Problem. Suppose that X is a Binomial random variable with $n = 100$ and unknown p. We want to test H_0: $p = 0.4$ against H_1: $p > 0.4$. We reject H_0 if $X \geq 48$. Use normal approximations throughout.

(a) Find the significance level (α) for this test.

(b) Suppose that the true value of p is 0.55. Find the power of the test.

Solution.
(a) Under H_0, $X \sim \text{Bin}(100, 0.4)$, so $\mu = 40$, $\sigma = \sqrt{100 \times 0.4 \times 0.6} = \sqrt{24}$.

$$\alpha = P(X \geq 48 \mid p = 0.4) \approx P\left(Z \geq \frac{47.5 - 40}{\sqrt{24}}\right) = P\left(Z \geq \frac{7.5}{4.899}\right) = P(Z \geq 1.531) \approx \boxed{0.063.}$$

(b) Under H_1, $X \sim \text{Bin}(100, 0.55)$, so $\mu = 55$, $\sigma = \sqrt{100 \times 0.55 \times 0.45} = \sqrt{24.75}$.

$$\text{Power} = P(X \geq 48 \mid p = 0.55) \approx P\left(Z \geq \frac{47.5 - 55}{\sqrt{24.75}}\right) = P\left(Z \geq \frac{-7.5}{4.975}\right) = P(Z \geq -1.508) \approx \boxed{0.934.}$$

Practice 8.7 — Assessing Normality (adapted from 2013 exam)

Problem. A machine produces metal rods. A random sample of 21 rods is selected and the diameter of each is measured. Assuming normality, a 95% confidence interval was constructed.

(a) How might you assess the assumption of normality? State two different methods. For each, indicate what would suggest normality and what would suggest non-normality.

Solution.

(a) Two methods to assess normality:

Method 1: Normal QQ-plot (or normal probability plot). Plot the data quantiles against the theoretical normal quantiles. If the points fall approximately on a straight line, this supports the normality assumption. Systematic curvature or deviations from linearity suggest non-normality.

Method 2: Formal hypothesis test (e.g., Shapiro–Wilk test, chi-squared goodness-of-fit

test). If you reject the null hypothesis (that the data come from a normal distribution), there is evidence of non-normality. If you fail to reject, the normality assumption is not contradicted.

Practice 8.8 — Sample Size for Power (adapted from 2013 exam)

Problem. Suppose $X_1, \ldots, X_n$ are i.i.d. $N(\mu, 1)$. We test H_0: $\mu = 10$ vs. H_a: $\mu \neq 10$ at the 5% level, rejecting if $\bar{X} > 10 + 1.96/\sqrt{n}$ or $\bar{X} < 10 - 1.96/\sqrt{n}$. If the true value of μ is 10.1, how large must n be for the test to have power at least 0.90?

Solution.

Under H_a: $\mu = 10.1$:

$$\text{Power} = P\left(\bar{X} > 10 + \frac{1.96}{\sqrt{n}} \;\middle|\; \mu = 10.1\right) + P\left(\bar{X} < 10 - \frac{1.96}{\sqrt{n}} \;\middle|\; \mu = 10.1\right).$$

The second term approaches 0 quickly. Focusing on the dominant term:

$$P\left(Z > \frac{10 + 1.96/\sqrt{n} - 10.1}{1/\sqrt{n}}\right) = P\big(Z > 1.96 - 0.1\sqrt{n}\big) \geq 0.90.$$

For $P(Z > c) = 0.90$, we need $c = -1.28$ (approximately). So:

$$1.96 - 0.1\sqrt{n} = -1.28 \implies 0.1\sqrt{n} = 3.24 \implies \sqrt{n} = 32.4 \implies n \approx 1050.$$

Therefore, n must be at least approximately $\boxed{1050}$ (the exact answer with full normal probabilities is $n = 1051$).

1.16 Law of Large Numbers — Informal Introduction

Setup: Let $X_1, X_2, \ldots$ be i.i.d. random variables with mean μ and variance σ^2. Define the sample mean:

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

Key observation (from CLT):

$$E(\bar{X}_n) = \mu, \qquad \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

As $n \to \infty$, $\text{Var}(\bar{X}_n) = \sigma^2/n \to 0$.

Since the variance of $\bar{X}_n$ shrinks to zero, the sample mean is becoming tightly concentrated around μ. This is the heart of the Law of Large Numbers.

So: $\bar{X}_n \to E(\bar{X}_n) = \mu$.

1.17 Weak Law of Large Numbers (WLLN)

Theorem: Weak Law of Large Numbers

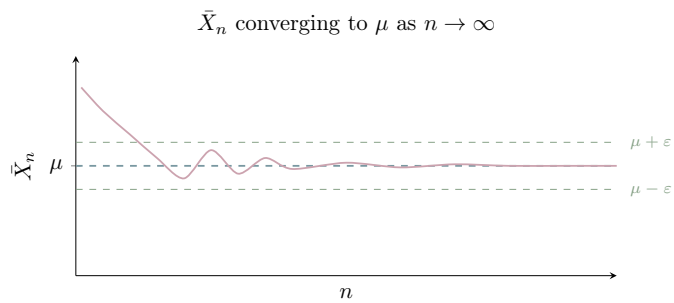
Let $X_1, X_2, \ldots$ be i.i.d. random variables with $E(X_i) = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Then for every $\varepsilon > 0$:

$$\lim_{n\to\infty} P\big(|\bar{X}_n - \mu| < \varepsilon\big) = 1.$$

That is, $\bar{X}_n$ **converges to μ in probability**.

What this means: No matter how small a tolerance ε you pick, as long as you take a large enough sample, the probability that $\bar{X}_n$ falls within ε of the true mean μ can be made as close to 1 as you like.

Proof: The WLLN is proved using Chebyshev's inequality (see Section 1.20).



1.18 Strong Law of Large Numbers (SLLN)

Theorem: Strong Law of Large Numbers

Under the same conditions as the WLLN:

$$P\Big(\lim_{n\to\infty} |\bar{X}_n - \mu| < \varepsilon\Big) = 1.$$

That is, $\bar{X}_n$ converges to μ **almost surely** (with probability 1).

Difference from the WLLN:

- **WLLN** (convergence in probability): For any fixed $\varepsilon > 0$, the probability that $\bar{X}_n$ is far from μ goes to 0. But different samples could still occasionally stray for a while.
- **SLLN** (almost sure convergence): The event that $\bar{X}_n \to \mu$ happens for sure — the limit of the sequence equals μ with probability 1. This is a stronger statement: it says the path $\bar{X}_n$ actually settles down to μ ("happens for sure").

Professor's remark: "For the exam, you mainly need to know the WLLN and how to prove it using Chebyshev. The SLLN is conceptual — know the difference, but you won't be asked to prove it. The key takeaway: more data = better estimate. The sample mean will converge to the population mean."

1.19 Markov's Inequality

Theorem: Markov's Inequality

If X is a **non-negative** random variable with $E(X) < \infty$ and $a > 0$, then:

$$P(X \geq a) \leq \frac{E(X)}{a}.$$

Intuitive interpretation: No more than $\frac{1}{5}$ of the population can have more than 5 times the average income. (Set $a = 5\mu$: $P(X \geq 5\mu) \leq \mu/(5\mu) = 1/5$.)

Proof (continuous case):

$$\begin{aligned} E(X) &= \int_0^\infty x f(x)\,dx = \int_0^a x f(x)\,dx + \int_a^\infty x f(x)\,dx \\ &\geq \int_a^\infty x f(x)\,dx \geq \int_a^\infty a\, f(x)\,dx = a\, P(X \geq a). \qquad \square \end{aligned}$$

Note: Markov's inequality is the foundational inequality — Chebyshev's inequality is derived from it.

Professor's remark: "Markov requires only $X \geq 0$ and $E(X) < \infty$ — very minimal assumptions. Chebyshev also needs the variance to exist, so it uses more information and gives tighter bounds. On the exam, always check: does the problem give you only a mean (use Markov) or both mean and variance (use Chebyshev)?"

1.20 Chebyshev's Inequality

Theorem: Chebyshev's Inequality

The probability that X assumes a value within k standard deviations of the mean is at least $1 - \frac{1}{k^2}$:

$$P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2}.$$

Example: For $k = 2$: $P(\mu - 2\sigma < X < \mu + 2\sigma) \geq 1 - \frac{1}{4} = \frac{3}{4}$.

Key point: Unlike the 68–95–99.7 rule (which requires normality), Chebyshev's inequality applies to **any distribution** with finite mean and variance. It is a weaker bound but universally applicable.

Proof (continuous case): See the uploaded proof notes. The key steps are:

1. Write $\sigma^2 = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\,dx$ and split into three regions.
2. Drop the middle integral (it is non-negative), keeping only the outer tails.
3. Use $(x-\mu)^2 \geq k^2\sigma^2$ on the tails to replace and simplify.
4. Divide by $k^2\sigma^2$ and complement to get the inequality.

Proof of the WLLN via Chebyshev's inequality:

Apply Chebyshev's inequality to $\bar{X}_n$, which has mean μ and variance σ^2/n:

$$P\big(|\bar{X}_n - \mu| \geq \varepsilon\big) \leq \frac{\mathrm{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \xrightarrow{n\to\infty} 0.$$

Therefore $P(|\bar{X}_n - \mu| < \varepsilon) \to 1$. $\square$

1.21 Sampling Distribution of S^2

Recall the sample variance:

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

1.21.1 Theorem: Chi-Squared Distribution of S^2

Theorem: Sampling Distribution of S^2

If S^2 is the sample variance of a random sample of size n taken from a **normal** distribution with variance σ^2, then:

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2} = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{\sigma^2} \sim \chi^2(n-1),$$

the chi-squared distribution with $(n-1)$ **degrees of freedom**.

Why $n-1$ degrees of freedom? We lose one degree of freedom because we estimated the mean μ with $\bar{X}$. The deviations $X_i - \bar{X}$ satisfy $\sum(X_i - \bar{X}) = 0$, so only $n-1$ are free to vary independently.

Professor's remark: "This result is extremely important — it is the basis for testing claims about population variance. The key thing to remember: this result **requires normality**. Unlike the CLT which works for any distribution, the chi-squared distribution of S^2 specifically needs the underlying population to be normal. Also remember that $\bar{X}$ and S^2 are independent when sampling from a normal distribution — this is a unique property of the normal."

1.21.2 Proof of the Chi-Squared Result

Step 1 — Decomposition identity: We start by writing:

$$\sum_{i=1}^{n}(X_i - \mu)^2 = \sum_{i=1}^{n}\left[(X_i - \bar{X}) + (\bar{X} - \mu)\right]^2.$$

Expanding:

$$= \sum_{i=1}^{n}(X_i - \bar{X})^2 + \sum_{i=1}^{n}(\bar{X} - \mu)^2 + 2(\bar{X} - \mu)\underbrace{\sum_{i=1}^{n}(X_i - \bar{X})}_{=0} = \sum_{i=1}^{n}(X_i - \bar{X})^2 + n(\bar{X} - \mu)^2.$$

Step 2 — Divide by σ^2: Substituting $(n-1)S^2$ for $\sum(X_i - \bar{X})^2$:

$$\frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2 = \frac{(n-1)S^2}{\sigma^2} + \frac{(\bar{X}-\mu)^2}{\sigma^2/n}.$$

Step 3 — Identify the pieces: Since $X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$:

- $\dfrac{X_i - \mu}{\sigma} \sim N(0,1)$, so $\left(\dfrac{X_i-\mu}{\sigma}\right)^2 \sim \chi^2(1)$.
- $\dfrac{1}{\sigma^2}\displaystyle\sum_{i=1}^{n}(X_i-\mu)^2 =$ sum of n independent $\chi^2(1)$ r.v.'s $\sim \chi^2(n)$.
- $\dfrac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$, so $\dfrac{(\bar{X}-\mu)^2}{\sigma^2/n} \sim \chi^2(1)$.

By the additivity of chi-squared (if $A \sim \chi^2(n)$ and $B \sim \chi^2(1)$ are independent and $A = C + B$, then $C \sim \chi^2(n-1)$):

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1). \qquad \square$$

*Additionally, this shows that $\bar{X}$ and S^2 are **independent** of each other when sampling from a normal population — a remarkable fact unique to the normal distribution.*

1.21.3 Chi-Squared Additivity Theorem

Theorem: Additivity of Chi-Squared

If $X_1 \sim \chi^2(\nu_1),\ X_2 \sim \chi^2(\nu_2), \ldots, X_k \sim \chi^2(\nu_k)$ are independent, then:

$$\sum_{i=1}^{k} X_i \sim \chi^2(\nu_1 + \nu_2 + \cdots + \nu_k).$$

This can be easily proved using the chi-squared moment generating function.

1.22 Summary: Sampling Distributions from a Normal Population

Key Results Summary (from the Board)

1. If $X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$ for $i = 1, \ldots, n$, then $\bar{X} \sim N\left(\mu, \dfrac{\sigma^2}{n}\right)$ **exactly**.

2. **CLT**: If $X_i \overset{\text{iid}}{\sim}$ any distribution with mean μ and variance σ^2, then $\bar{X} \overset{\text{approx}}{\sim} N\left(\mu, \dfrac{\sigma^2}{n}\right)$ for $n \geq 30$.

3. If $X_{1i} \sim N(\mu_1, \sigma_1^2)$ (sample of n_1) and $X_{2i} \sim N(\mu_2, \sigma_2^2)$ (sample of n_2) independently, then:

$$\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2,\ \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right).$$

1.23 Worked Example: Chi-Squared Test for Variance

Example — Car Battery Lifetimes

Problem. A manufacturer guarantees its car batteries will last, on average, 3 years with $\sigma \leq 1$ year. Five batteries are tested and found to have lifetimes (in years): 1.9, 2.4, 3.0, 3.5, 4.2. Is the claim $\sigma^2 \leq 1$ correct?

Given: $n = 5$, so $n - 1 = 4$ degrees of freedom. Assume $\sigma^2 = 1$ (null hypothesis).

Step 1 — Compute S^2:

$$\bar{x} = \frac{1.9 + 2.4 + 3.0 + 3.5 + 4.2}{5} = \frac{15.0}{5} = 3.0.$$

$$S^2 = \frac{(1.9-3)^2 + (2.4-3)^2 + (3.0-3)^2 + (3.5-3)^2 + (4.2-3)^2}{4} = \frac{1.21 + 0.36 + 0 + 0.25 + 1.44}{4} = \frac{3.26}{4} = 0.815.$$

Step 2 — Compute the test statistic:

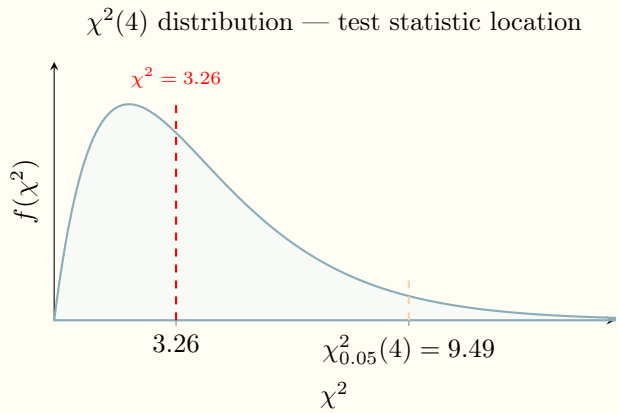
Assume $\sigma^2 = 1$ and lifetime is normally distributed.

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2} = \frac{4 \times 0.815}{1} = 3.26 \ \sim\ \chi^2(4).$$

Step 3 — Find the p-value:

We want $P(\chi^2(4) > 3.26)$. From the $\chi^2(4)$ table, $P(\chi^2 > 0.5153) \approx$ high. Since 3.26 is well within the central mass of a $\chi^2(4)$ distribution, we **fail to reject** H_0.

Conclusion: The data do not provide sufficient evidence to reject the manufacturer's claim that $\sigma^2 \leq 1$.



1.24 Practice Problems

Practice 8.9 — LLN and Chebyshev

Problem. Let $X_1, X_2, \ldots$ be i.i.d. with mean $\mu = 5$ and variance $\sigma^2 = 4$.

(a) Using Chebyshev's inequality, find a lower bound on $P(3 < \bar{X}_{100} < 7)$.

(b) What does the WLLN tell you about $\bar{X}_n$ as $n \to \infty$?

(c) If in addition the $X_i \sim N(5, 4)$, compute the exact probability $P(3 < \bar{X}_{100} < 7)$.

Solution.
(a) $\bar{X}_{100}$ has mean 5 and $\mathrm{Var}(\bar{X}_{100}) = 4/100 = 0.04$, so $\mathrm{SD}(\bar{X}_{100}) = 0.2$.
The interval $(3, 7)$ is $\mu \pm 2$, i.e. $\mu \pm k\sigma_{\bar{X}}$ where $k = 2/0.2 = 10$.
By Chebyshev: $P(3 < \bar{X}_{100} < 7) \geq 1 - 1/10^2 = 1 - 0.01 = \mathbf{0.99}$.
(b) The WLLN says $\bar{X}_n \to 5$ in probability; for any $\varepsilon > 0$, $P(|\bar{X}_n - 5| < \varepsilon) \to 1$.
(c) $\bar{X}_{100} \sim N(5, 0.04)$ exactly (since X_i normal). Standardize: $z = (3-5)/0.2 = -10$ and $z = (7-5)/0.2 = 10$. $P(3 < \bar{X}_{100} < 7) = P(-10 < Z < 10) \approx \mathbf{1.0000}$.

Practice 8.10 — Markov and Chebyshev Comparison

Problem. $X \geq 0$ has mean 4.

(a) Use Markov's inequality to bound $P(X \geq 20)$.

(b) If additionally $\mathrm{Var}(X) = 2$, use Chebyshev's inequality to bound $P(|X - 4| \geq 4)$.

(c) Which inequality gives a tighter bound in (b)? Why?

Solution.
(a) $P(X \geq 20) \leq E(X)/20 = 4/20 = \mathbf{0.2}$.
(b) Chebyshev: $P(|X - 4| \geq 4) \leq \mathrm{Var}(X)/4^2 = 2/16 = \mathbf{0.125}$.

(c) Chebyshev gives a tighter bound (0.125 vs. Markov applied to $|X-4|^2$ would give the same; Chebyshev uses variance information that Markov does not).

Practice 8.11 — Sampling Distribution of S^2 (adapted from 2012 exam)

Problem. A random sample of size $n = 10$ is taken from a normal population with variance $\sigma^2 = 2$. Find $P(S^2 > 3.5)$.

Solution.
$\dfrac{(n-1)S^2}{\sigma^2} = \dfrac{9S^2}{2} \sim \chi^2(9)$.
We want:

$$P(S^2 > 3.5) = P\left(\frac{9S^2}{2} > \frac{9 \times 3.5}{2}\right) = P(\chi^2(9) > 15.75).$$

From the $\chi^2(9)$ table: $\chi^2_{0.05}(9) = 16.92$ and $\chi^2_{0.10}(9) = 14.68$.
Since $14.68 < 15.75 < 16.92$, we have $0.05 < P(\chi^2(9) > 15.75) < 0.10$.
$\therefore P(S^2 > 3.5) \approx \mathbf{0.07}$ (between 5% and 10%).

Practice 8.12 — Chi-Squared Additivity (adapted from past exam)

Problem. Suppose $X_1 \sim \chi^2(3)$ and $X_2 \sim \chi^2(5)$ are independent.

(a) What is the distribution of $X_1 + X_2$?

(b) Find $P(X_1 + X_2 > 15.507)$.

(c) Find $E(X_1 + X_2)$ and $\mathrm{Var}(X_1 + X_2)$.

Solution.
(a) By additivity: $X_1 + X_2 \sim \chi^2(3+5) = \boldsymbol{\chi^2(8)}$.
(b) $P(\chi^2(8) > 15.507) = 0.05$ (this is the standard 5% critical value for 8 d.f.).
(c) $E(\chi^2(\nu)) = \nu$ and $\mathrm{Var}(\chi^2(\nu)) = 2\nu$:

$$E(X_1 + X_2) = 3 + 5 = \mathbf{8}, \qquad \mathrm{Var}(X_1 + X_2) = 2(3) + 2(5) = \mathbf{16}.$$

1.25 March 11: Review of Key Concepts

Before introducing new sampling distributions, we review the foundational concepts and results from earlier chapters that Chapter 8 builds upon.

Population: The entire group of individuals that you are interested in.

Sample: A subset of the population.

Parameter: A numerical characteristic of the *population* (e.g. μ, σ^2). Parameters are fixed but usually unknown.

Statistic: A numerical quantity calculated from the *sample* (e.g. $\bar{X}$, S^2). Statistics are random variab/Users/rhyls/Desktop/school/Winter26/MIE286/Lecture notes/final lecture notes/plot.pngles because they de-

pend on which sample is drawn.

Sampling Distribution: The probability distribution of a statistic. Since a statistic is a random variable, it has a distribution — this distribution is what we call the sampling distribution.

Law of Large Numbers: As the sample size $n \to \infty$, the sample mean $\bar{X}$ converges to the population mean μ.

Central Limit Theorem (CLT): For large n, the distribution of $\bar{X}$ is approximately normal regardless of the underlying population distribution.

Key results from previous chapters (all assume X_i are independent):

1. If $X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$, then
$$\bar{X} \sim N\left(\mu,\ \frac{\sigma^2}{n}\right).$$
Here S^2 is the sample variance.

2. If $X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$, then
$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1).$$

3. If $X_{1i} \sim N(\mu_1, \sigma_1^2)$ and $X_{2i} \sim N(\mu_2, \sigma_2^2)$ are independent samples, then
$$\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2,\ \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right).$$

Note: For results (1) and (3), you do not actually need the populations to be normally distributed. By the Central Limit Theorem, if n is large enough, $\bar{X}$ will still be approximately normally distributed. It is not exact, but it would be approximately normally distributed.

1.26 Motivation: Why Do We Need New Distributions?

Recall that when we made inferences about the population mean μ from a normal population, we used the standard normal distribution via the Z-statistic:
$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$$

For this, we needed a known value of σ^2. But in practice, if you do not know the population mean μ (which is the whole reason you are doing inference), then you are also not going to know σ^2. You need an estimate for σ^2, and that estimate is the sample variance S^2.

When we replace σ with S, we can no longer use the normal distribution. Instead, we use the **t-distribution**.

1.27 The t-Distribution

1.27.1 Definition of the t-Distribution

Theorem: Definition of the t-Distribution

Let $Z \sim N(0,1)$ and $V \sim \chi^2(\nu)$, and suppose Z and V are **independent**.
Then the random variable

$$T = \frac{Z}{\sqrt{V/\nu}}$$

has pdf

$$h(t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{\pi\nu}} \left(1 + \frac{t^2}{\nu}\right)^{-(\nu+1)/2}, \qquad -\infty < t < \infty.$$

This is called the **t-distribution with ν degrees of freedom**, written $T \sim t(\nu)$.

This pdf can be derived through the transformation techniques from Chapter 7, but one can show that this is the pdf for this random variable. You do not need to memorize the pdf — it involves gamma functions just like the chi-squared pdf.

1.27.2 The One-Sample t-Statistic

Using the definition above (a combination of two independent random variables Z and V), we can show the following key result:

Theorem: Sampling Distribution of the t-Statistic

Let $X_1, \ldots, X_n$ be independent random variables, all normally distributed with mean μ and variance σ^2 (i.e. $X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$). Then

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1).$$

Proof:

Define

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1), \qquad V = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1).$$

From a result proved in earlier chapters, $\bar{X}$ and S^2 are **independent** when the population is normal. Therefore Z and V are independent.

Now form the ratio:

$$T = \frac{Z}{\sqrt{V/(n-1)}} = \frac{\dfrac{\bar{X} - \mu}{\sigma/\sqrt{n}}}{\sqrt{\dfrac{(n-1)S^2}{\sigma^2} \cdot \dfrac{1}{n-1}}} = \frac{\dfrac{\bar{X} - \mu}{\sigma/\sqrt{n}}}{\sqrt{\dfrac{S^2}{\sigma^2}}} = \frac{\dfrac{\bar{X} - \mu}{\sigma/\sqrt{n}}}{\dfrac{S}{\sigma}} = \frac{\bar{X} - \mu}{S/\sqrt{n}}.$$

Since $Z \sim N(0,1)$, $V \sim \chi^2(n-1)$, and they are independent, the definition of the t-distribution gives

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1). \qquad \square$$

Note: The key step is that $\bar{X}$ and S^2 are independent (a fact specific to normal populations). This is why the σ cancels in the numerator and denominator, leaving us with the t-statistic that does not depend on σ.

1.27.3 Historical Note: The "Student" Distribution

History: Why "Student's t"?

The t-distribution is also called the **Student distribution**. It was discovered by William Sealy Gosset, who worked for the Guinness brewery in the early 1900s. His job was to ensure that production had low variability from batch to batch.

He was working with **small sample sizes** and did not know σ^2. If he had had a very large sample size, his estimate S^2 would have been very close to σ^2, and it would have been fine to use the standard normal distribution. But because he had to work with small samples, he needed a distribution that accounted for the extra uncertainty introduced by estimating σ.

However, Guinness did not allow employees to publish company findings. So Gosset published under the **pseudonym** (alias) "Student." That is how the distribution came to be called "Student's t-distribution."

Ronald Fisher, one of the founders of experimental design and statistics, helped Gosset with some of the mathematical proofs in the publication. Fisher also developed the F-distribution. There was also Karl Pearson (of the Pearson correlation coefficient). These are the "fathers of statistics" who worked in the early 1900s. Interestingly, there was some rivalry: Pearson's journal *Biometrika* (still published today) never allowed Fisher to publish in it, despite Fisher being such a major figure in statistics.

1.27.4 Properties of the t-Distribution

Useful tips:

1. **When n is small, S is unreliable.** If we knew σ, then $Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$. The T-statistic does exactly the same thing except it replaces σ with S, the sample standard deviation.

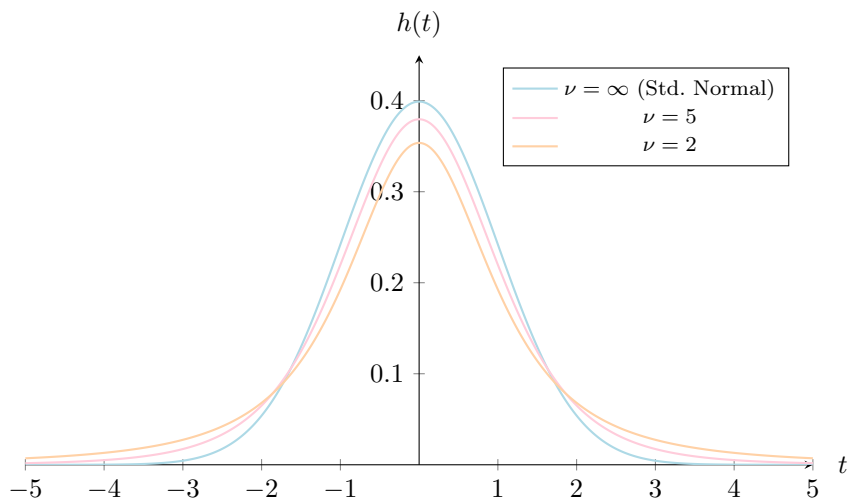
   Now, σ is a constant, but S is a random variable. If you select 20 observations, you get one value of S^2. If someone else selects a different 20 observations, they get a *different* S^2. If yet another person selects 20, there is yet another S^2. This variability in S^2 is captured by the t-distribution having heavier tails than the standard normal.

   The Z-statistic has mean 0 and variance 1. The t-distribution also has expected value 0, but because S is random (not constant like σ), it has **larger variability** — i.e., $\mathrm{Var}(T) > 1$. Thus, the t-distribution has **wider tails** than the standard normal.

2. **As $\nu \to \infty$, $t(\nu) \to N(0,1)$.** When $\nu = \infty$, the t-distribution is exactly the standard normal. This makes sense: if you have many observations in your sample, then S will be very close to σ. The larger

the sample, the less variable S is. So the smallest variability corresponds to the standard normal, and as the sample size increases, the t-distribution gets closer and closer to it.

3. **Practical rule of thumb ($n \geq 30$):** It turns out that if n is around 30, the t-distribution is practically the same as the standard normal. So 30 is the "magic number" — it works both for the CLT and for the t-distribution convergence.



The plot above shows that as ν decreases, the t-distribution has heavier tails (more spread out) compared to the standard normal. All three curves are symmetric about 0. As ν increases, the t-distribution converges to the standard normal.

1.27.5 Assumptions and Robustness

When Can You Use the t-Distribution?

- The formal assumption is that the **population is normally distributed**. This is *not* a CLT result — the derivation requires normality.
- However, the t-distribution is **robust**: even if the population is not exactly normal, but is **symmetric** with a large enough sample size, the t-distribution will still work well in practice.
- If the population distribution is **non-symmetric**, you generally need a larger sample size for the t-distribution to be a reliable approximation.
- We use the t-distribution for making inferences about the **population mean** μ, or for **comparing two population means**.

1.27.6 Reading the t-Distribution Table (Table A.4)

The t-distribution table is structured differently from the standard normal table:

- In the standard normal table, the entries are probabilities. In the t-table, the entries are the **t-values** (critical values).

- The rows correspond to the **degrees of freedom** $\nu = n - 1$.
- The columns correspond to the **upper-tail probability** α. The entry t_α is the value such that $P(T > t_\alpha) = \alpha$.
- For example, with $\nu = 3$ and $\alpha = 0.30$: the entry 0.277 means that a t-value of 0.277 has a 40% probability to its right... wait, let me re-read — actually, $t_{0.30}$ with $\nu = 3$ is 0.584. The idea is: $P(T > 0.584) = 0.30$.

Important observation: Look at the table after $\nu = 30$. For a given α, the t-values change very little beyond this point. And at $\nu = \infty$, the values are *exactly* the standard normal critical values. For instance, $t_{0.025}$ at $\nu = \infty$ is 1.960, which is precisely $z_{0.025}$. You can verify: 1.96 appears both in the t-table at $\nu = \infty$ and in the standard normal table. So in general, we say the t-distribution beyond 30 degrees of freedom is effectively the standard normal.

1.28 The F-Distribution

We use the F-distribution when we compare two different variances from two different populations, or when we compare more than two population means simultaneously (ANOVA, which we will cover later).

1.28.1 Definition of the F-Distribution

Theorem: Definition of the F-Distribution

Let $U \sim \chi^2(\nu_1)$ and $V \sim \chi^2(\nu_2)$, and suppose U and V are **independent**.
Then the random variable

$$F = \frac{U/\nu_1}{V/\nu_2}$$

has pdf

$$h(f) = \frac{\Gamma\left(\frac{\nu_1+\nu_2}{2}\right)}{\Gamma\left(\frac{\nu_1}{2}\right)\Gamma\left(\frac{\nu_2}{2}\right)} \left(\frac{\nu_1}{\nu_2}\right)^{\nu_1/2} \cdot \frac{f^{(\nu_1/2)-1}}{\left(1 + \frac{\nu_1 f}{\nu_2}\right)^{(\nu_1+\nu_2)/2}}, \qquad f > 0.$$

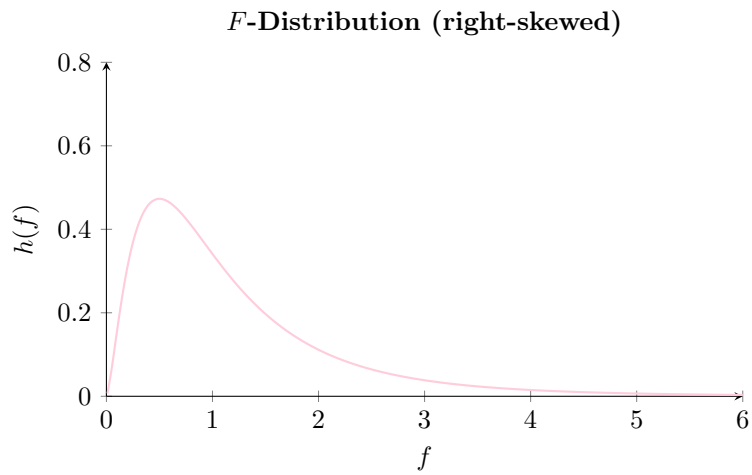
This is called the **F-distribution with ν_1 and ν_2 degrees of freedom**, written $F \sim F(\nu_1, \nu_2)$.

Like the t-distribution pdf, you do not need to memorize this — it involves gamma functions. This is the pdf of the random variable that has the F-distribution.

1.28.2 Properties of the F-Distribution

- The F-distribution is **right-skewed** (positively skewed) and takes only **positive values** ($f > 0$). Note: the t-distribution is symmetric, the chi-squared distribution is right-skewed, and the F-distribution is also right-skewed.
- It is **not symmetric**.

- The degrees of freedom ν_1 (numerator) and ν_2 (denominator) are **not interchangeable**: $F(\nu_1, \nu_2) \neq F(\nu_2, \nu_1)$ in general.



Reciprocal property: Imagine rather than computing U divided by V, you could also compute V divided by U. That should also be an F-distribution, just with swapped degrees of freedom. Formally: if $F \sim F(\nu_1, \nu_2)$, then

$$\frac{1}{F} \sim F(\nu_2, \nu_1).$$

This is just changing what is in the numerator and what is in the denominator.

1.28.3 The F-Statistic for Comparing Two Variances

Theorem: F-Statistic for Two Sample Variances

If S_1^2 and S_2^2 are the variances of independent random samples of sizes n_1 and n_2, taken from normal populations with variances σ_1^2 and σ_2^2 respectively, then

$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1 - 1,\ n_2 - 1).$$

Proof:

We can control this by looking at these random variables, which we know are chi-squared distributed from result (2) in our review:

$$U = \frac{(n_1 - 1)S_1^2}{\sigma_1^2} \sim \chi^2(n_1 - 1), \qquad V = \frac{(n_2 - 1)S_2^2}{\sigma_2^2} \sim \chi^2(n_2 - 1).$$

Since the two samples are independent, U and V are independent. Then if we take U divided by $(n_1 - 1)$

over V divided by $(n_2 - 1)$:

$$F = \frac{U/(n_1-1)}{V/(n_2-1)} = \frac{\dfrac{(n_1-1)S_1^2}{\sigma_1^2} \cdot \dfrac{1}{n_1-1}}{\dfrac{(n_2-1)S_2^2}{\sigma_2^2} \cdot \dfrac{1}{n_2-1}} = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1-1, n_2-1). \qquad \square$$

Note: If we assume $\sigma_1^2 = \sigma_2^2$ (i.e. we are testing whether two populations have equal variances), the F-statistic simplifies to $F = S_1^2/S_2^2$.

Reciprocal property for table look-up: Since tables typically only give upper-tail critical values, for lower-tail values we use:

$$f_{1-\alpha}(\nu_1, \nu_2) = \frac{1}{f_\alpha(\nu_2, \nu_1)}.$$

1.28.4 Reading the F-Distribution Table

The F-distribution has two degrees of freedom, so we need more tables than for the t-distribution:

- Each table corresponds to a specific α value (e.g. $\alpha = 0.05$, $\alpha = 0.01$, $\alpha = 0.025$). These tend to be our thresholds in decision-making.
- The rows are ν_2 (denominator degrees of freedom) and the columns are ν_1 (numerator degrees of freedom).
- The entry is the critical value $f_\alpha(\nu_1, \nu_2)$ such that $P(F > f_\alpha) = \alpha$.
- For example, in the $\alpha = 0.05$ table, the entry at $\nu_1 = 1$ and $\nu_2 = 8$ gives the F-value that has 5% probability to its right for an $F(1, 8)$ distribution.
- If you need more precise values (or degrees of freedom not in the table), you can use R to compute F-quantiles directly.

We use the F-distribution when we compare two different sample groups that potentially come from different populations. For example, if you have 20 samples from one population and 30 from another, your degrees of freedom would be $\nu_1 = 19$ and $\nu_2 = 29$.

1.29 Summary: When to Use What

Situation	**Distribution**	**Statistic**
Inference about μ, σ known	Normal (Z)	$Z = \dfrac{X-\mu}{\sigma/\sqrt{n}}$
Inference about μ, σ unknown	t-distribution	$T = \dfrac{X-\mu}{S/\sqrt{n}}$
Inference about σ^2	Chi-squared	$\dfrac{(n-1)S^2}{\sigma^2}$
Comparing two variances	F-distribution	$\dfrac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}$

So the two new distributions we learned today — t and F — are what we use when we do not assume σ^2 is known. The t-distribution is for inference about the population mean. The F-distribution is for comparing population variances (and later, for comparing more than two population means via ANOVA).

Exercise 8.24

Problem: Given $n = 36$, $\mu_{\bar{X}} = 40$, $\sigma_{\bar{X}} = 2/6 = 1/3$, find $P\left(\sum_{i=1}^{36} X_i > 1458\right)$.
Solution: $z = (40.5 - 40)/(1/3) = 1.5$. $P(\sum X_i > 1458) = P(\bar{X} > 40.5) = P(Z > 1.5) = 1 - 0.9332 = \boxed{0.0668}$.

Exercise 8.26

Problem: $n = 64$, $\mu_{\bar{X}} = 3.2$, $\sigma_{\bar{X}} = 1.6/8 = 0.2$. Find (a) $P(\bar{X} < 2.7)$; (b) $P(\bar{X} > 3.5)$; (c) $P(3.2 < \bar{X} < 3.4)$.
Solution: (a) $z = (2.7 - 3.2)/0.2 = -2.5$, $P(\bar{X} < 2.7) = P(Z < -2.5) = \boxed{0.0062}$.
(b) $z = (3.5 - 3.2)/0.2 = 1.5$, $P(\bar{X} > 3.5) = P(Z > 1.5) = 1 - 0.9332 = \boxed{0.0668}$.
(c) $z_1 = (3.2 - 3.2)/0.2 = 0$, $z_2 = (3.4 - 3.2)/0.2 = 1.0$. $P(3.2 < \bar{X} < 3.4) = P(0 < Z < 1.0) = 0.8413 - 0.5000 = \boxed{0.3413}$.

Exercise 8.30

Problem: $\mu_1 - \mu_2 = 0$, $\sigma_{\bar{X}_1 - \bar{X}_2} = 50\sqrt{1/32 + 1/50} = 11.319$. Find (a) $P(|\bar{X}_1 - \bar{X}_2| > 20)$; (b) $P(-10 < \bar{X}_1 - \bar{X}_2 < -5) + P(5 < \bar{X}_1 - \bar{X}_2 < 10)$.
Solution: (a) $z_1 = -20/11.319 = -1.77$, $z_2 = 20/11.319 = 1.77$. $P(|\bar{X}_1 - \bar{X}_2| > 20) = 2P(Z < -1.77) = 2(0.0384) = \boxed{0.0768}$.
(b) $z_1 = 5/11.319 = 0.44$, $z_2 = 10/11.319 = 0.88$. $P(-10 < \bar{X}_1 - \bar{X}_2 < -5) + P(5 < \bar{X}_1 - \bar{X}_2 < 10) = 2P(0.44 < Z < 0.88) = 2(0.8106 - 0.6700) = \boxed{0.2812}$.

Exercise 8.44

Problem: Find the following t-values from the table.
Solution: (a) $t_{0.025}$ with $\nu = 15$: $\boxed{2.145}$.
(b) $-t_{0.10}$ with $\nu = 10$: $\boxed{-1.372}$.
(c) $-t_{0.005}$ with $\nu = 7$: $\boxed{-3.499}$.

Exercise 8.45

Problem: Find the following probabilities involving the t-distribution.
Solution: (a) $P(T < 2.365) = 1 - 0.025 = \boxed{0.975}$.
(b) $P(T > 1.318) = \boxed{0.10}$.
(c) $P(T < 2.179) = 0.975$ and $P(T < -1.356) = P(T > 1.356) = 0.10$. Therefore $P(-1.356 < T < 2.179) = 0.975 - 0.010 = \boxed{0.875}$.
(d) $P(T > -2.567) = 1 - P(T > 2.567) = 1 - 0.01 = \boxed{0.99}$.

Exercise 8.47

Problem: From the t-table (Table A.4), find the following values when $\nu = 23$.

Solution: (a) $t_{0.025} = 2.069$ when $\nu = 23$. Therefore $-t_{0.025} = -2.069$, and the total area to the left of $t = k$ is $0.025 + 0.965 = 0.990$. Hence $k = t_{0.01} = \boxed{2.500}$.

(b) $t_{0.005} = 2.807$ when $\nu = 23$. The total area to the right of $t = k$ is $0.095 + 0.005 = 0.10$. Hence $k = t_{0.10} = \boxed{1.319}$.

(c) $t_{0.05} = \boxed{1.714}$ for 23 degrees of freedom.

Exercise 8.48

Problem: Given $t_{0.025} = 2.131$ for $\nu = 15$, test whether the observed t-value falls within the acceptance region.

Solution: $t = \frac{27.5-30}{5/4} = -2.00$. Since $-2.131 < -2.00 < 2.131$, the test statistic falls between $-t_{0.025}$ and $t_{0.025}$, so the claim is $\boxed{\text{valid}}$.

Exercise 8.49

Problem: A random sample of 9 observations gives $\bar{x} = 24$, $s = 4.1$. Test whether $\mu > 20$.

Solution: $t = \frac{24-20}{4.1/3} = 2.927$. With 8 degrees of freedom, $t_{0.01} = 2.896$. Since $t = 2.927 > 2.896$, we reject H_0. Conclusion: no, $\mu > 20$. $\boxed{\text{Reject } H_0}$.

Chapter 2

Chapter 9: One- and Two-Sample Estimation

2.1 Point Estimation

2.1.1 Introduction to Point Estimation

Point estimation is the foundation of statistical inference. When we collect sample data, we need to make educated guesses about the true population parameters we cannot observe directly.

Definition: Point Estimate

A point estimate of some population parameter θ is a single value $\hat{\theta}$ of a statistic that we compute from our sample data. Common examples include $\bar{X}$ to estimate μ, or the sample proportion $\hat{p}$ to estimate p.

Think of a point estimate as your best single guess for an unknown population parameter. If we want to estimate the average height of all UofT students, we'd calculate the mean height of a sample and use that as our point estimate. But because we're guessing from a sample, there's always some uncertainty involved—we wouldn't expect our guess to be exactly right.

The challenge with point estimation is that there are often multiple reasonable choices for an estimator. For example, to estimate the population mean μ, we could use $\bar{X}$ (the sample mean), or the median, or even some weighted average. How do we choose which estimator is better?

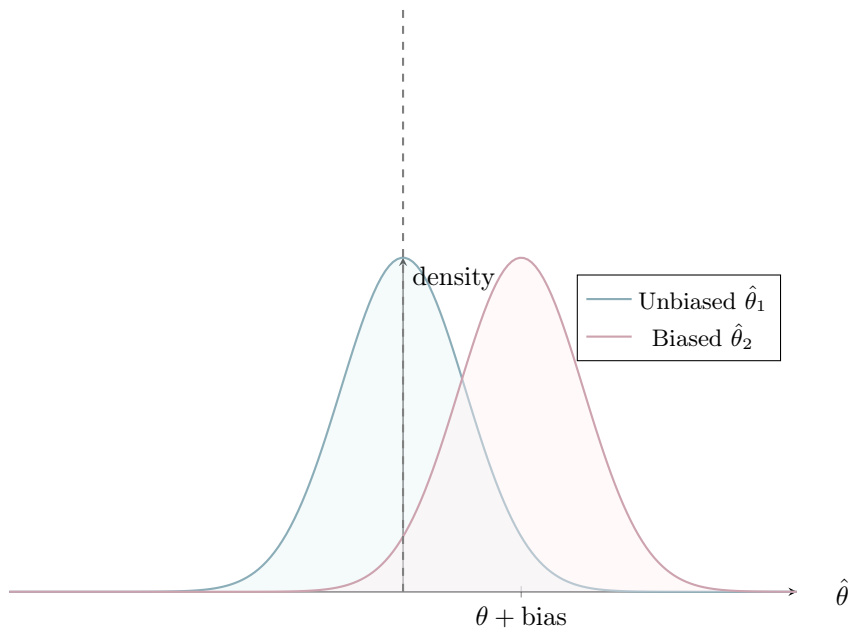
2.1.2 Unbiased Estimators

The first criterion we use to evaluate estimators is whether they're *unbiased*—that is, whether they tend to hit the target on average across all possible samples.

Definition: Unbiased Estimator

A statistic $\hat{\theta}$ is said to be an <u>unbiased estimator</u> of the parameter θ if $E(\hat{\theta}) = \theta$. In other words, the expected value of the estimator equals the true parameter.

An unbiased estimator is like throwing darts at a bullseye: on average, your darts cluster around the center, even if each individual throw isn't perfectly centered. The mean of all your throws would be right at the target. A biased estimator, on the other hand, would consistently aim slightly to the left of center.



In the graph above, the blue curve (unbiased estimator) is centered right at the true parameter θ. The pink curve (biased estimator) is centered somewhere else—its expected value misses the target.

2.1.2.1 Theorem: $\bar{X}$ is Unbiased for μ

Theorem: Unbiasedness of Sample Mean

If $X_1, X_2, \ldots, X_n$ is a random sample from a distribution with mean μ, then $\bar{X}$ is an unbiased estimator of μ. That is, $E(\bar{X}) = \mu$.

Proof:

$$\begin{aligned} E(\bar{X}) &= E\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right) \\ &= \frac{1}{n}E(X_1) + \frac{1}{n}E(X_2) + \cdots + \frac{1}{n}E(X_n) \\ &= \frac{1}{n}\mu + \frac{1}{n}\mu + \cdots + \frac{1}{n}\mu = \frac{n\mu}{n} = \mu \end{aligned}$$

The sample mean is unbiased because of the linearity of expectation. We can split up the expected value of the sum into the sum of expected values, and since each observation comes from the same distribution with

mean μ, everything cancels out to give us exactly μ.

2.1.2.2 Theorem: S^2 is Unbiased for σ^2

Theorem: Unbiasedness of Sample Variance

$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$ is an unbiased estimator of σ^2. That is, $E(S^2) = \sigma^2$.

Proof:

First, expand the sum of squared deviations. The trick is to add and subtract μ:

$$\begin{aligned}\sum_{i=1}^{n}(X_i - \bar{X})^2 &= \sum_{i=1}^{n}[(X_i - \mu) - (\bar{X} - \mu)]^2 \\ &= \sum_{i=1}^{n}(X_i - \mu)^2 - 2(\bar{X} - \mu)\sum_{i=1}^{n}(X_i - \mu) + n(\bar{X} - \mu)^2\end{aligned}$$

Since $\sum_{i=1}^{n}(X_i - \mu) = n(\bar{X} - \mu)$:

$$\sum_{i=1}^{n}(X_i - \bar{X})^2 = \sum_{i=1}^{n}(X_i - \mu)^2 - n(\bar{X} - \mu)^2$$

Taking expectations:

$$\begin{aligned}E(S^2) &= \frac{1}{n-1}\left[\sum_{i=1}^{n} E(X_i - \mu)^2 - n\,E(\bar{X} - \mu)^2\right] \\ &= \frac{1}{n-1}\left[n\sigma^2 - n \cdot \frac{\sigma^2}{n}\right] = \frac{1}{n-1}\left[n\sigma^2 - \sigma^2\right] = \sigma^2\end{aligned}$$

where we used $\sigma^2_{X_i} = \sigma^2$ and $\sigma^2_{\bar{X}} = \sigma^2/n$.

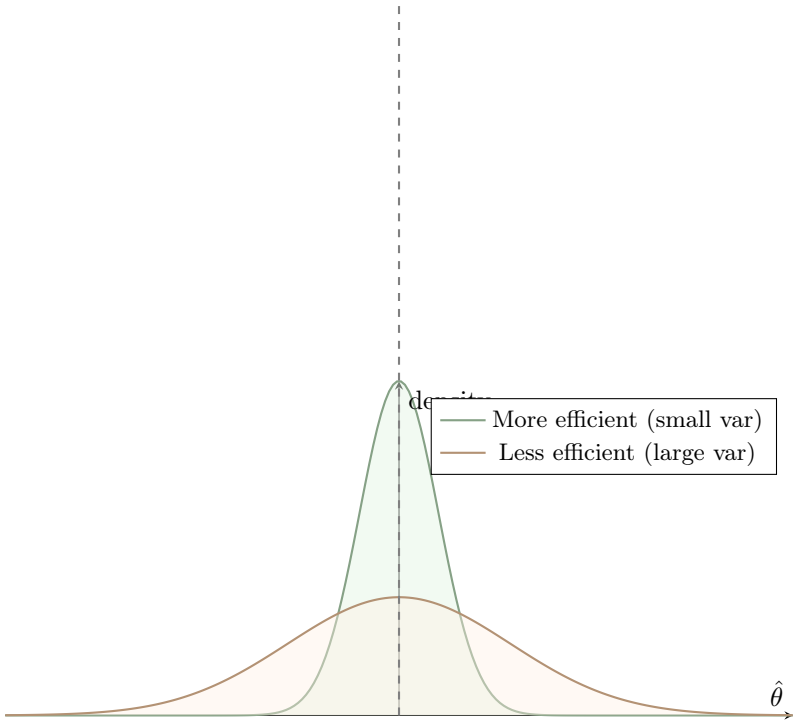
This is why we divide by $n-1$ instead of n in the sample variance formula! If we divided by n, the formula would be biased (it would systematically underestimate the true variance). The $n-1$ adjustment, called Bessel's correction, compensates for the fact that we're measuring deviations from $\bar{X}$ rather than from the true μ.

2.1.3 Comparing Estimators: Efficiency

Once we've established that an estimator is unbiased, we ask: among all unbiased estimators, which one should we prefer?

Definition: Most Efficient Estimator

If $\hat{\theta}_1$ and $\hat{\theta}_2$ are two unbiased estimators of θ, we prefer the one with the <u>smaller variance</u>. If we consider *all* possible unbiased estimators of θ, the one with the smallest variance is called the <u>most efficient estimator</u>.



Both estimators above are unbiased (centered at θ), but the green one is much more efficient—its estimates cluster tightly around the true value, giving you more reliable results each time you sample.

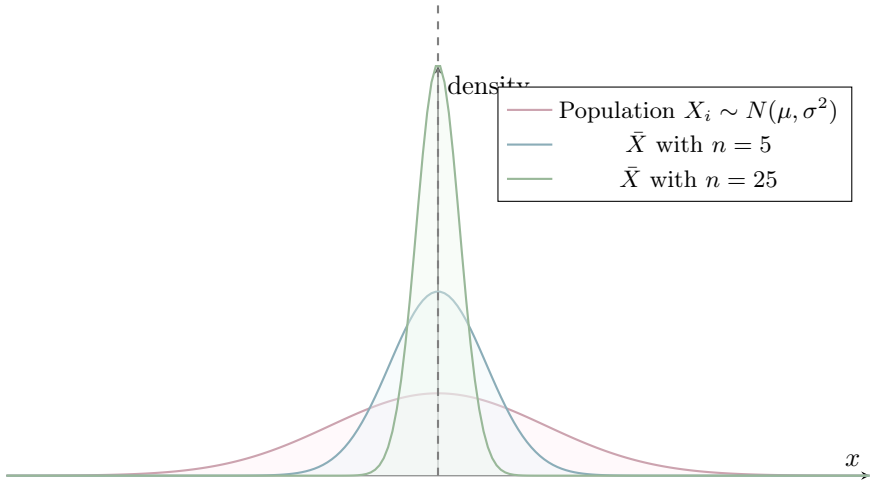
2.1.4 Standard Error

Key Concept: Standard Error

The standard error (SE) of a statistic is the standard deviation of its sampling distribution. For $X_i \overset{iid}{\sim} N(\mu, \sigma^2)$:

$$\bar{X} \sim N\left(\mu,\ \frac{\sigma^2}{n}\right)$$

- σ is the standard deviation of each individual observation X_i.
- $\frac{\sigma}{\sqrt{n}}$ is the standard error of $\bar{X}$.



This graph shows the key idea: as sample size n increases, the sampling distribution of $\bar{X}$ gets taller and narrower. The population distribution (pink, wide) shows how individual observations vary. But sample means vary much less—and the more data you collect, the more precise your estimate becomes. Specifically, to cut the standard error in half, you need to quadruple your sample size (because of the $\sqrt{n}$ in the denominator).

2.2 Interval Estimation and Confidence Intervals

2.2.1 Introduction to Confidence Intervals

Point estimates are useful, but they don't tell us about uncertainty. A confidence interval addresses this by providing a *range* of plausible values for the parameter.

Definition: Confidence Interval

A confidence interval (CI) reports a range of plausible values for θ:

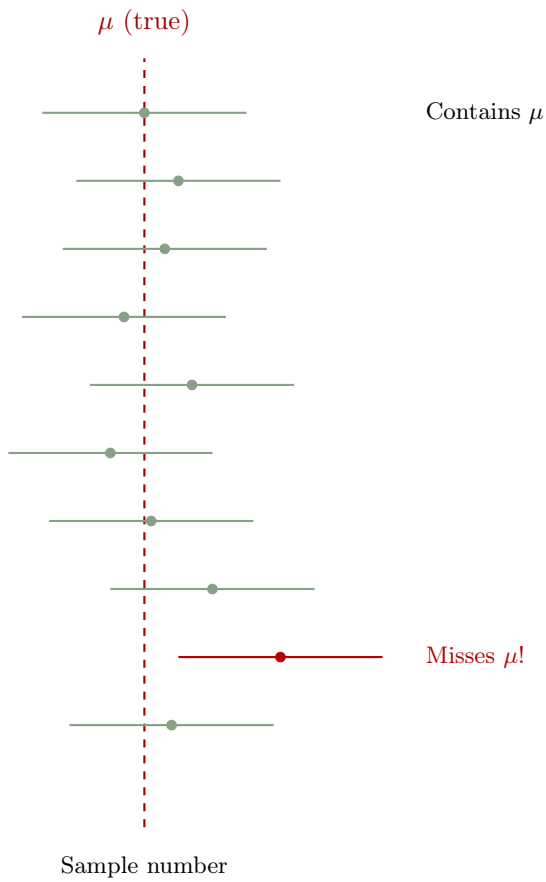
$$P(\hat{\theta}_L < \theta < \hat{\theta}_U) = 1 - \alpha, \quad 0 < \alpha < 1$$

where $\hat{\theta}_L$ and $\hat{\theta}_U$ are random variables (from the sample), and $1 - \alpha$ is the confidence level.

A confidence interval is like saying: "Based on my sample, I'm 95% confident the true value lies somewhere between A and B." The bounds are random because they depend on which sample we draw. The confidence level refers to the long-run frequency: if we repeated our sampling procedure many times, about 95% of the intervals we construct would contain the true parameter.

Important Interpretation

It is **incorrect** to say "there is a 95% probability that θ lies in our computed interval." Once we compute a specific interval, θ is either in it or not. The 95% refers to the *procedure's* success rate over many repetitions.



Each horizontal line above represents a confidence interval from a different sample. The dot is $\bar{x}$ for that sample. Most intervals (green) capture the true μ (dashed red line), but occasionally one (red) misses. With a 95% confidence level, about 1 in 20 intervals will miss.

2.2.2 Trade-off Between Precision and Confidence

There is a tradeoff: higher confidence $\Rightarrow$ wider (less precise) intervals. We want a *tight* CI with a *high* confidence level, but we can't fully have both.

2.2.3 CI for μ When σ is Known

Theorem: CI for μ (Known σ)

If $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ with σ known, then the $100(1-\alpha)\%$ CI for μ is:

$$\bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

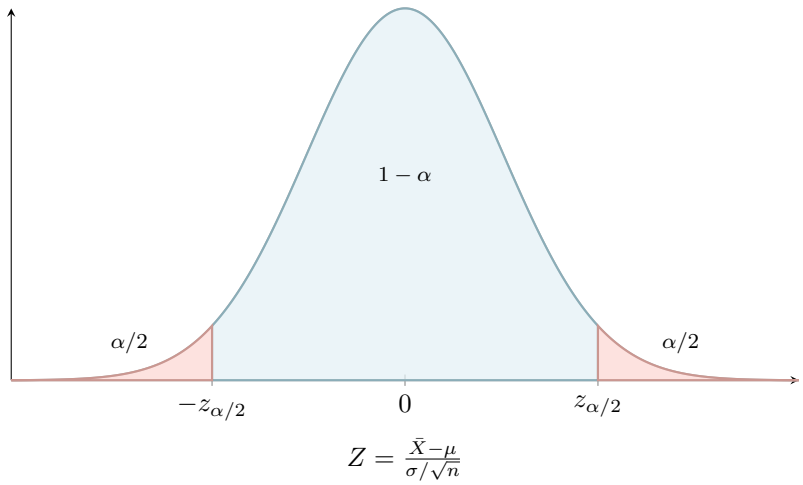
where $z_{\alpha/2}$ satisfies $P(Z > z_{\alpha/2}) = \alpha/2$.

Derivation: Since $\bar{X} \sim N(\mu, \sigma^2/n)$, we standardize:

$$P\left(-z_{\alpha/2} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < z_{\alpha/2}\right) = 1 - \alpha$$

Rearranging for μ:

$$P\left(\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$



The blue central region contains probability $1-\alpha$. The pink tails each contain $\alpha/2$. For a 95% CI, $\alpha = 0.05$, and the critical value is $z_{0.025} = 1.96$.

2.2.3.1 Common Critical Values

Confidence Level	α	$z_{\alpha/2}$
90%	0.10	1.645
95%	0.05	1.960
99%	0.01	2.576

2.2.3.2 Worked Example: Lightbulb Lifetime

Lightbulb Lifespan Estimation

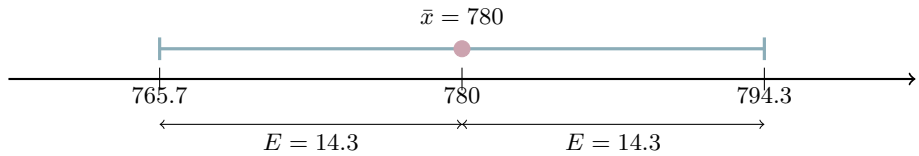
A random sample of $n = 30$ lightbulbs has $\bar{x} = 780$ hours. Assume $\sigma = 40$ hours. Find the 95% CI for μ.

Solution: $z_{0.025} = 1.96$, and the margin of error is:

$$E = 1.96 \times \frac{40}{\sqrt{30}} = 1.96 \times 7.30 = 14.31 \text{ hours}$$

$$95\% \text{ CI}: \quad 780 \pm 14.31 = \boxed{(765.7,\ 794.3) \text{ hours}}$$

We're 95% confident the true mean lifespan is between 765.7 and 794.3 hours.



2.2.4 Sample Size for Specified Error

Key Concept: Sample Size Determination

We can be $100(1-\alpha)\%$ confident that the error $|\bar{X} - \mu|$ will not exceed e if:

$$n \geq \left(\frac{z_{\alpha/2} \cdot \sigma}{e}\right)^2$$

Round up to the nearest integer.

This formula shows three things: (1) smaller error e requires larger n; (2) more variable populations (larger σ) require larger n; (3) higher confidence (larger z) requires larger n. All of these are intuitive!

Determining Required Sample Size — Lightbulbs

How large a sample is required if we want to be 95% confident that our estimate of μ is off by less than 10 hours? Assume $\sigma = 40$ hours.

Solution:

$$n \geq \left(\frac{1.96 \times 40}{10}\right)^2 = (7.84)^2 = 61.47$$

Therefore $n = 62$ or more. $\boxed{n = 62}$

2.2.5 One-Sided Confidence Intervals

CIs need not be two-sided. Sometimes we only care about a bound in one direction.

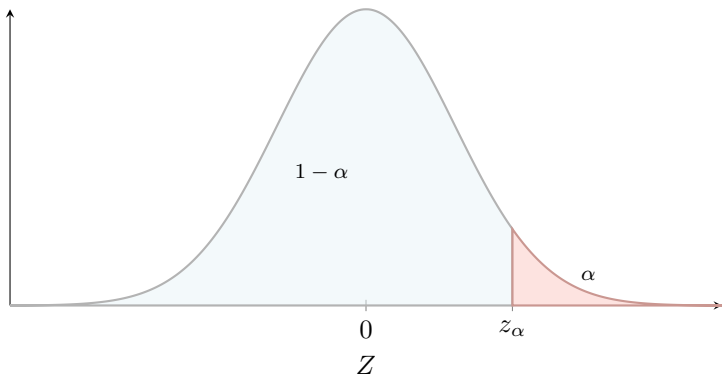
Key Concept: One-Sided Confidence Intervals

Upper one-sided CI: $\left(-\infty,\ \bar{x} + z_\alpha \cdot \frac{\sigma}{\sqrt{n}}\right)$

Lower one-sided CI: $\left(\bar{x} - z_\alpha \cdot \frac{\sigma}{\sqrt{n}},\ \infty\right)$

Note: use z_α (not $z_{\alpha/2}$) since the entire α is in one tail.

Upper One-Sided: All of α in the right tail



For a one-sided CI, all of the α probability goes into one tail instead of being split between two. This means the critical value is z_α rather than $z_{\alpha/2}$, which makes the bound slightly tighter in the direction we care about. A one-sided CI is useful when you only care about one direction—for example, "what's the minimum lifespan we can guarantee?"

2.3 CI for μ When σ is Unknown

In most real-world situations, σ is unknown and must be estimated by S.

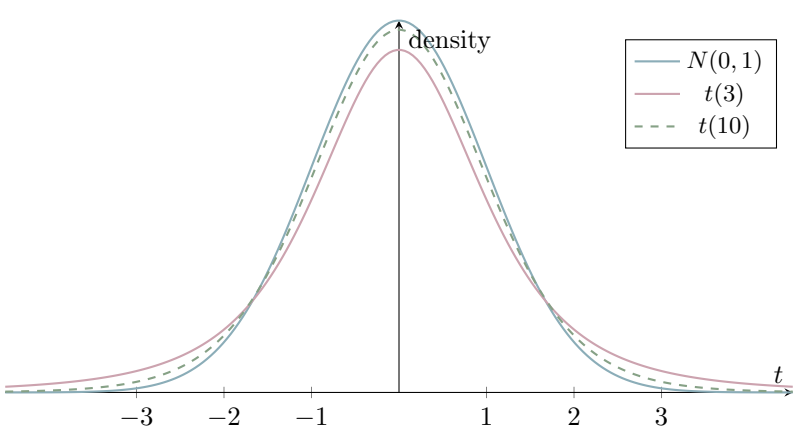
Theorem: CI for μ (Unknown σ)

If $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ with σ unknown, then:

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$$

and the $100(1-\alpha)\%$ CI for μ is:

$$\bar{x} \pm t_{\alpha/2,\, n-1} \cdot \frac{s}{\sqrt{n}}$$



The t-distribution has heavier tails than the normal, accounting for the extra uncertainty from estimating σ. With only $df = 3$, the tails are much fatter (pink curve). As df increases (green dashed curve, $df = 10$), the t-distribution approaches the standard normal (blue). For $n > 30$, they're virtually identical.

2.4 CI for Population Proportion

When data are binary (yes/no), we estimate p using $\hat{p}$.

Theorem: CI for Proportion p

For large n (where $n\hat{p} \geq 5$ and $n\hat{q} \geq 5$), by the CLT:

$$\hat{p} = \frac{\sum X_i}{n} \approx N\left(p, \frac{pq}{n}\right)$$

The $100(1-\alpha)\%$ CI for p is:

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}}$$

where $\hat{q} = 1 - \hat{p}$.

Each observation X_i is Bernoulli(p), so $\bar{X} = \hat{p}$. By the CLT, the sample proportion is approximately normal for large n. We plug in $\hat{p}$ and $\hat{q}$ into the standard error since the true p is unknown.

HBO Subscription Proportion

In a random sample of $n = 500$ families in a particular city, 340 subscribe to HBO. Find the 95% CI for the actual proportion p.

Solution:

$$\hat{p} = \frac{340}{500} = 0.68, \quad \hat{q} = 0.32, \quad z_{0.025} = 1.96$$

$$SE = \sqrt{\frac{(0.68)(0.32)}{500}} = \sqrt{0.000435} = 0.0209$$

$$\hat{p} \pm 1.96 \times 0.0209 = 0.68 \pm 0.041$$

$$\boxed{0.64 < p < 0.72}$$

We're 95% confident that between 64% and 72% of families in this city subscribe to HBO.

2.5 Prediction Intervals

A prediction interval predicts where a *single new observation* X_0 will fall, rather than estimating the population mean.

Prediction Interval for a New Observation

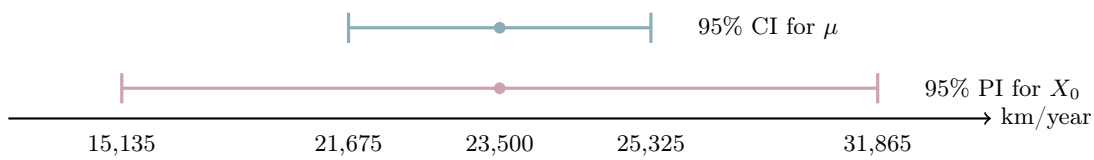
Since $\bar{X} \sim N(\mu, \sigma^2/n)$ and $X_0 \sim N(\mu, \sigma^2)$ independently:

$$X_0 - \bar{X} \sim N\left(0,\ \sigma^2 + \frac{\sigma^2}{n}\right) = N\left(0,\ \sigma^2\left(1 + \frac{1}{n}\right)\right)$$

σ **known:** $\bar{x} \pm z_{\alpha/2}\, \sigma \sqrt{1 + \frac{1}{n}}$

σ **unknown:** $\bar{x} \pm t_{\alpha/2,\, n-1}\, s \sqrt{1 + \frac{1}{n}}$

The key factor is $\sqrt{1 + 1/n}$ instead of $\sqrt{1/n}$. The "1" accounts for the variability of a single observation around μ, and the "$1/n$" accounts for estimation uncertainty in $\bar{X}$. Even with $n \to \infty$, a prediction interval stays wide because individual observations always vary!



Car Driving Distance — CI vs. Prediction Interval

A random sample of $n = 20$ car owners: $\bar{x} = 23{,}500$ km/year, $s = 3{,}900$ km. Assume normal.

(a) 95% CI for μ:

$$t_{0.025,\,19} = 2.093, \quad SE = \frac{3900}{\sqrt{20}} = 872$$

$$23{,}500 \pm 2.093 \times 872 = 23{,}500 \pm 1,825$$

$$\boxed{(21{,}675,\ 25{,}325)\ \text{km/year}}$$

(b) 95% Prediction Interval for a new car owner:

$$23{,}500 \pm 2.093 \times 3{,}900 \times \sqrt{1 + \tfrac{1}{20}} = 23{,}500 \pm 2.093 \times 3{,}900 \times 1.025$$

$$= 23{,}500 \pm 8{,}365$$

$$\boxed{(15{,}135,\ 31{,}865)\ \text{km/year}}$$

*The prediction interval is much wider! We're fairly confident about the **average**, but predicting one individual's driving is far more uncertain.*

2.6 Two-Sample Estimation

Often we want to compare two populations by estimating $\mu_1 - \mu_2$.

2.6.1 Case 1: Two Samples with Known Variances

CI for $\mu_1 - \mu_2$ (Known Variances)

If independent samples are from $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$:

$$\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2,\ \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$$

The $100(1-\alpha)\%$ CI for $\mu_1 - \mu_2$:

$$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

The key insight: variances add for independent random variables. The variance of the difference $\bar{X}_1 - \bar{X}_2$ is the sum of the individual variances.

2.6.2 Case 2: Unknown but Equal Variances (Pooled)

CI for $\mu_1 - \mu_2$ (Unknown Equal Variances)

When $\sigma_1^2 = \sigma_2^2 = \sigma^2$ (unknown), we combine both samples to estimate σ^2:

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2} \qquad \text{(pooled variance)}$$

The $100(1-\alpha)\%$ CI:

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2,\, n_1+n_2-2} \cdot S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Pooling is like combining information from both samples to get a better estimate of the shared variability. The pooled variance is a weighted average of S_1^2 and S_2^2, with each sample weighted by its degrees of freedom.

2.6.3 Case 3: Unknown and Unequal Variances (Welch)

CI for $\mu_1 - \mu_2$ (Unknown Unequal Variances — Welch)

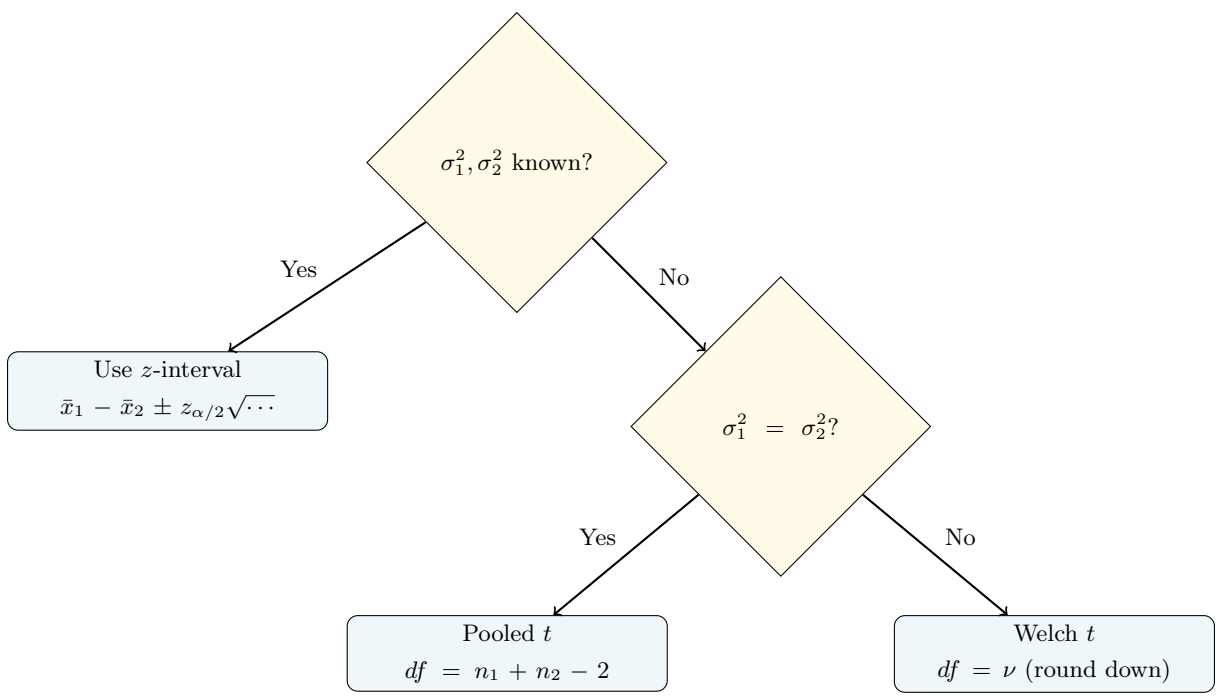
$$T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \sim t(\nu)$$

where ν is approximated by (Welch–Satterthwaite):

$$\nu = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{(S_1^2/n_1)^2}{n_1-1} + \frac{(S_2^2/n_2)^2}{n_2-1}} \quad \text{(round down to nearest integer)}$$

CI: $(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2,\,\nu}\sqrt{\dfrac{S_1^2}{n_1} + \dfrac{S_2^2}{n_2}}$

This is the most general and safest approach—it doesn't assume equal variances. The Welch–Satterthwaite formula for degrees of freedom looks scary but is just a way to account for the fact that two separate variance estimates contribute to our uncertainty.



2.6.3.1 Worked Example: Tire Brand Comparison

Two-Sample CI: Tire Durability

A cab company tests 12 tires each from Brand A and B until they wear out:

- Brand A: $\bar{x}_A = 36{,}300$ km, Brand B: $\bar{x}_B = 38{,}100$ km
- $\bar{x}_A - \bar{x}_B = -1{,}800$ km

Compute 95% CI for $\mu_A - \mu_B$ under each scenario.

(a) Known variances: $\sigma_A = 5000$, $\sigma_B = 6100$.

$$-1800 \pm 1.96\sqrt{\frac{5000^2}{12} + \frac{6100^2}{12}} = -1800 \pm 4463 = \boxed{(-6263,\ 2663)}$$

(b) Unknown equal variances:

$$S_p = \sqrt{\frac{11 \times S_A^2 + 11 \times S_B^2}{22}}, \quad df = 22, \quad t_{0.025,22} = 2.074$$

$$\boxed{(-6522,\ 2922)}$$

(c) Unknown unequal variances (Welch):

$$\nu \approx 21, \quad t_{0.025,21} = 2.080$$

$$\boxed{(-6536,\ 2936)}$$

Decision: Since 0 is in all three intervals, we **fail to reject** that $\mu_A = \mu_B$.

2.7 Paired Observations

When observations come in pairs (before/after, left/right, same subject measured twice), we analyze the *differences* directly.

Paired Observations Setup

For n pairs (X_{1i}, X_{2i}), define:

$$D_i = X_{1i} - X_{2i}, \quad i = 1, \ldots, n$$

$$\mathrm{Var}(D_i) = \sigma_1^2 + \sigma_2^2 - 2\,\mathrm{Cov}(X_{1i}, X_{2i})$$

Since within-pair correlation is typically positive, pairing **reduces variance**!
The $100(1-\alpha)\%$ CI for $\mu_D = \mu_1 - \mu_2$:

$$\bar{d} \pm t_{\alpha/2,\, n-1} \cdot \frac{s_d}{\sqrt{n}}$$

where n is the number of *pairs*.

Pairing is powerful because it controls for individual differences. Testing the same students before and after a course removes the effect of "some students are just better at English"—what's left is purely the course effect. This typically leads to smaller variance and narrower CIs than treating the groups as independent.

Paired Analysis: TOEFL Improvement

A college offers English courses. 20 students take TOEFL before and after. Differences $D_i = \text{Post} - \text{Pre}$:

Student	1	2	3	$\cdots$	20
Pre	30	28	31	$\cdots$	29
Post	29	30	32	$\cdots$	32
D_i	-1	2	1	$\cdots$	3

Solution:

$$\bar{d} = 1.45, \quad s_d = 3.203, \quad n = 20, \quad t_{0.025,\,19} = 2.02$$

$$1.45 \pm 2.02 \times \frac{3.203}{\sqrt{20}} = 1.45 \pm 1.45$$

$$\boxed{(-0.049,\ 2.999)}$$

The interval just barely contains 0, so the evidence for improvement is marginal. The course may help, but we can't confidently claim it does at the 95% level.

2.8 Confidence Intervals for Variance

2.8.1 CI for a Single Population Variance (σ^2)

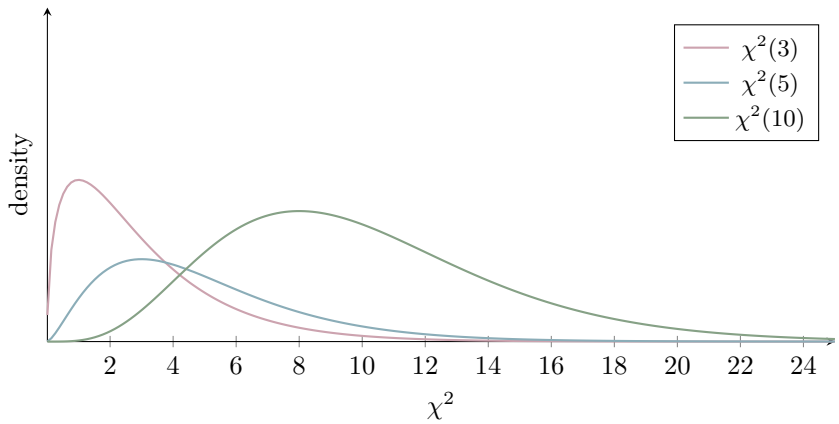
CI for σ^2 Using Chi-Square

For $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$:

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$$

The $100(1-\alpha)\%$ CI for σ^2:

$$\left(\frac{(n-1)s^2}{\chi^2_{\alpha/2,\, n-1}}, \quad \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\, n-1}} \right)$$



The chi-squared distribution is right-skewed (not symmetric like the normal or t). That's why the CI for σ^2 is asymmetric—the upper bound extends further than the lower bound. The larger the degrees of freedom, the more symmetric the χ^2 distribution becomes.

CI for Tire Variance

Brand A tires: $n_A = 12$, $s_A^2 = 5000^2 = 25{,}000{,}000$ km^2. Find 95% CI for σ_A^2.
Solution: From χ^2 tables with $df = 11$: $\chi^2_{0.025,11} = 21.920$, $\chi^2_{0.975,11} = 3.816$.

$$\text{Lower} = \frac{11 \times 25{,}000{,}000}{21.920} = 12{,}546{,}000, \quad \text{Upper} = \frac{11 \times 25{,}000{,}000}{3.816} = 72{,}067{,}000$$

$$\boxed{\sigma_A^2 \in (12{,}546{,}000,\ 72{,}067{,}000)\ \text{km}^2}$$

In standard deviation units: $\sigma_A \in (3{,}542,\ 8{,}490)$ km.
Notice how wide this interval is! Estimating variance from small samples is inherently imprecise.

2.8.2 CI for Ratio of Two Variances (σ_1^2/σ_2^2)

CI for Variance Ratio Using F-Distribution

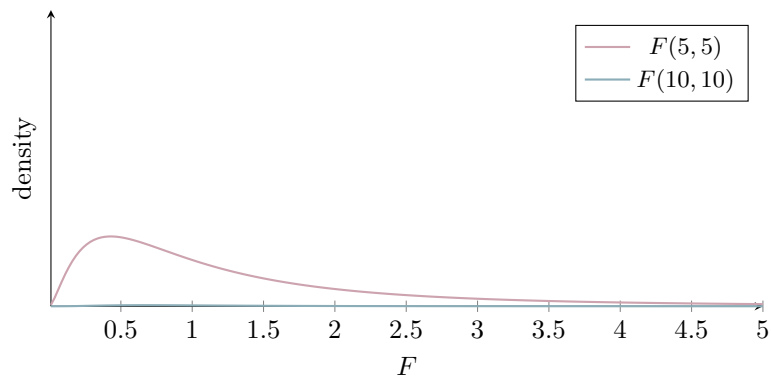
For independent samples from $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$:

$$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1 - 1,\ n_2 - 1)$$

The $100(1-\alpha)\%$ CI for σ_1^2/σ_2^2:

$$\left(\frac{S_1^2}{S_2^2} \cdot \frac{1}{F_{\alpha/2,\, n_1-1,\, n_2-1}},\ \ \frac{S_1^2}{S_2^2} \cdot F_{\alpha/2,\, n_2-1,\, n_1-1}\right)$$

If the CI for σ_1^2/σ_2^2 contains 1, we cannot conclude the variances differ. If the interval is entirely above 1, population 1 is more variable; if entirely below 1, population 2 is more variable.



Comparing Tire Variance: Are $\sigma_A^2 = \sigma_B^2$?

Brand A: $s_A^2 = 5000^2$, Brand B: $s_B^2 = 6100^2$, both $n = 12$.

Solution:

$$\frac{s_A^2}{s_B^2} = \frac{5000^2}{6100^2} = 0.672$$

From F-tables: $F_{0.025,\,11,\,11} = 3.474$.

$$\text{Lower} = \frac{0.672}{3.474} = 0.193, \quad \text{Upper} = 0.672 \times 3.474 = 2.335$$

$$\boxed{(0.193,\ 2.335)}$$

Since 1 is inside the interval, we **fail to reject** $\sigma_A^2 = \sigma_B^2$.

2.9 Maximum Likelihood Estimation (MLE)

MLE is a powerful general approach to finding point estimates.

Key Concept: Likelihood Function

Given a random sample $X_1, \ldots, X_n$ from a distribution with parameter θ, the likelihood function is:

$$L(\theta \mid x_1, \ldots, x_n) = f(x_1 \mid \theta) \cdot f(x_2 \mid \theta) \cdots f(x_n \mid \theta) = \prod_{i=1}^{n} f(x_i \mid \theta)$$

This is the joint density of the data, viewed as a function of θ.

Definition: Maximum Likelihood Estimator

The MLE $\hat{\theta}$ is the value of θ that *maximizes* $L(\theta)$—the parameter value that makes our observed data most probable. In practice, we maximize the log-likelihood $\ell(\theta) = \ln L(\theta)$.

The intuition is beautiful: "What value of the parameter makes the data I actually observed most likely?" Rather than relying on moment equations or ad hoc arguments, MLE asks a single, principled question.

2.9.1 Worked Example: Bernoulli MLE

MLE for Assembly Line Defects

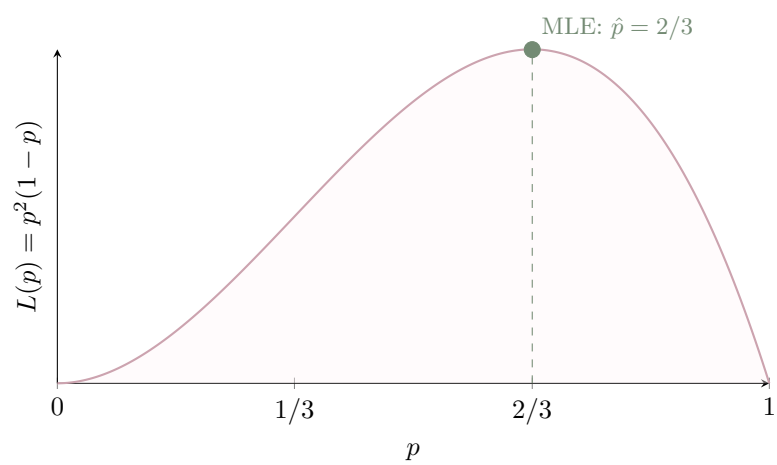
Three items from an assembly line are inspected. We get 2 nondefective followed by 1 defective. Estimate p, the proportion of nondefective items, using MLE.

Solution:

$$L(p) = p \cdot p \cdot (1-p) = p^2(1-p) = p^2 - p^3$$

$$\frac{dL}{dp} = 2p - 3p^2 = p(2-3p) = 0 \implies p = 0 \text{ or } p = \tfrac{2}{3}$$

Since $L(0) = 0$, the maximum is at $\boxed{\hat{p} = 2/3}$.



The graph shows the likelihood function—its peak at $p = 2/3$ is the MLE. This is the value that makes observing "success, success, failure" most probable.

2.9.2 MLE for the Normal Distribution

MLE for Normal: Deriving $\hat{\mu}$ and $\hat{\sigma}^2$

Consider a random sample from $N(\mu, \sigma^2)$. Find the MLEs of μ and σ^2.

Solution:

The likelihood is:

$$L(\mu, \sigma^2) = \frac{1}{(2\pi)^{n/2}(\sigma^2)^{n/2}} \exp\left[-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2\right]$$

The log-likelihood is:

$$\ell = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2$$

Setting partial derivatives to zero:

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i - \mu) = 0 \implies \boxed{\hat{\mu} = \bar{x}}$$

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(x_i - \mu)^2 = 0 \implies \boxed{\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Check: second derivatives are < 0, confirming a maximum.

Important: The MLE of σ^2 divides by n, not $n - 1$! So the MLE is *biased*. However, for large n, the bias is negligible.

2.9.3 Properties of MLEs

Properties of Maximum Likelihood Estimators

1. **Consistency:** $\hat{\theta} \xrightarrow{p} \theta$ as $n \to \infty$
2. **Asymptotic normality:** For large n, $\hat{\theta}$ is approximately normal
3. **Asymptotic efficiency:** MLEs achieve the lowest possible variance among consistent estimators
4. **Invariance:** If $\hat{\theta}$ is the MLE of θ, then $g(\hat{\theta})$ is the MLE of $g(\theta)$

These properties explain why MLEs are so widely used in practice: they converge to the truth, follow a known distribution for inference, are as precise as possible (for large samples), and the invariance property means you get MLEs of transformed parameters "for free."

2.10 Chapter 9 Summary: Quick Reference

Scenario	**Confidence Interval**
μ, σ known	$\bar{x} \pm z_{\alpha/2}\dfrac{\sigma}{\sqrt{n}}$
μ, σ unknown	$\bar{x} \pm t_{\alpha/2,n-1}\dfrac{s}{\sqrt{n}}$
Proportion p	$\hat{p} \pm z_{\alpha/2}\sqrt{\dfrac{\hat{p}\hat{q}}{n}}$
Prediction for X_0 (σ unknown)	$\bar{x} \pm t_{\alpha/2,n-1}\, s\sqrt{1+\dfrac{1}{n}}$
$\mu_1 - \mu_2$, σ's known	$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2}\sqrt{\dfrac{\sigma_1^2}{n_1}+\dfrac{\sigma_2^2}{n_2}}$
$\mu_1 - \mu_2$, equal σ's	$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2,n_1+n_2-2}\, S_p\sqrt{\dfrac{1}{n_1}+\dfrac{1}{n_2}}$
$\mu_1 - \mu_2$, unequal σ's	$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2,\nu}\sqrt{\dfrac{S_1^2}{n_1}+\dfrac{S_2^2}{n_2}}$
Paired: μ_D	$\bar{d} \pm t_{\alpha/2,n-1}\dfrac{s_d}{\sqrt{n}}$
σ^2	$\left(\dfrac{(n-1)s^2}{\chi^2_{\alpha/2}}, \dfrac{(n-1)s^2}{\chi^2_{1-\alpha/2}}\right)$
σ_1^2/σ_2^2	$\left(\dfrac{s_1^2}{s_2^2}\cdot\dfrac{1}{F_{\alpha/2}}, \dfrac{s_1^2}{s_2^2}\cdot F_{\alpha/2}\right)$
Sample size	$n \geq \left(\dfrac{z_{\alpha/2}\sigma}{e}\right)^2$